Communication Frictions, Managerial Expectations, and Firm Dynamics *

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Haoran Wang †

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Abstract

This paper studies how imperfect communication within firms shapes managerial expectations, input choices, and aggregate productivity. I develop a dynamic model of team production in which capital and labor managers share the firm's objective but observe segmented information about productivity. The model yields a novel empirical test using expectation data: investment forecasts underreact to hiring when capital managers fail to incorporate labor managers' private information. Using managerial forecast data of US public firms, I find that investment forecasts simultaneously underreact to hiring and overreact to investment, a pattern inconsistent with common-information models. Quantitatively, communication friction substantially weakens investment–hiring comovement and generates capital–labor-ratio dispersion equal to about 39% of its observed magnitude. The resulting MRPK and MRPN dispersions equal 27% and 19% of their observed counterparts, and the associated aggregate TFP loss is approximately 5% relative to the frictionless allocation. This productivity loss is limited relative to the marginal-product dispersion because segmented information induces a negative covariance between MRPK and MRPN.

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†Department of Economics, University of Maryland, College Park. Email: hw2688@umd.edu

1 Introduction

A large empirical literature establishes that management and organizational practices are important drivers of firm productivity (Bloom and Van Reenen, 2007; Bloom et al., 2013, 2019). However, less is known about which aspects of management matter and through what channels they affect firms' decisions and aggregate allocative efficiency. This paper studies a specific channel, linking how firms organize their internal information flows to their input decisions and aggregate productivity through managerial expectations. Using firm expectation data, I document that investment forecasts simultaneously overreact to past investment but underreact to past hiring. This opposite-sign pattern provides evidence against a common-information benchmark and supports a model of segmented information within firms. I show that this information-segmentation channel is quantitatively important: it generates substantial variations in the capital-labor ratio and dispersion in marginal revenue products, leading to a sizable 5% TFP loss relative to the frictionless allocation.

The analysis is motivated by strong evidence of imperfect communication among decision-makers within firms (Mian, 2006; Giroud, 2013; Malenko, 2022; Impink et al., 2024). Theoretically, I develop a dynamic team production framework in which each firm is modeled as a team consisting of a capital manager and a labor manager. The managers choose their respective inputs under uncertainty about the firm's productivity but share the objective of maximizing the firm's value. Imperfect communication is modeled by allowing each manager to receive a private signal about the firm's productivity, making the information sets of managers non-nested and segmented within the firm. The managers form expectations and make investment and hiring decisions based on these segmented information sets.

The model yields a novel empirical test for information segmentation. I show that the communication friction leads the rational investment forecast error of the capital manager to be positively predicted by past hiring.[1] The intuition is that, when the capital manager makes forecasts under segmented information, he is uncertain about the private signals the labor manager receives and the hiring decision she makes. As a result, his one-period-ahead rational forecasts underreact to hiring decisions. The actual investment, on the other hand, incorporates information revealed by the previous period's hiring, making the investment forecast error positively predicted by past hiring. This "cross-decision" predictability is impossible under common information: without information segmentation, the labor manager's hiring decision is included in the capital manager's information set when he forms expecta-

[1] Here, I define the rational investment forecast error as the difference between actual investment and the rational one-period-ahead forecast of investment.

tions, making it impossible for his rational forecast errors of investment to be predicted by past hiring.

Because reported forecasts in the data may also reflect behavioral expectations that deviate from rational expectations, cross-decision predictability alone is not sufficient to identify information segmentation. To address this empirical challenge, I decompose the cross-decision predictability into behavioral and rational components analytically. I show that under common information, the cross-decision predictability is solely driven by the behavioral component, which has the same sign as the predictability of investment forecast error by past investment (which I refer to as the "own-decision" predictability). This is because, under common information, if capital and labor are complements in the revenue function, investment and hiring move in the same direction in response to the same news about productivity. As a result, the behavioral overreaction or underreaction to the news makes the forecast error predictable by past investment and past hiring in the same direction. By contrast, communication friction breaks the tie between the signs of own- and cross-decision predictability, since the rational component of investment forecast error can be positively predicted by past hiring. Hence, the own- and cross-decision predictability showing opposite signs is inconsistent with the common-information benchmark and consistent with segmented information.

I test this prediction using a dataset of U.S. public firms that merges self-reported expectations of capital expenditure from IBES with realized outcomes from Compustat. In the data, investment forecasts display exactly this opposite-sign pattern: the investment forecast errors are negatively predicted by past investment but simultaneously positively predicted by past hiring. In other words, investment forecasts simultaneously overreact to investment but underreact to hiring. The opposite signs are inconsistent with the common-information benchmark and are not readily generated by the alternative common-information mechanisms considered in the paper, including non-convex adjustment costs, time-to-build and empirically plausible non-neutral technologies. The evidence therefore supports a model in which capital and labor decisions are based on segmented information.

I next ask whether communication friction is quantitatively relevant for firm dynamics and allocative efficiency. To do so, I extend the model to incorporate time-to-build in capital, over-extrapolative beliefs, and Hsieh–Klenow-style reduced-form distortions. The model is calibrated to match the own- and cross-decision predictabilities and standard moments of input adjustment, the responsiveness of input choices to productivity, and marginal-product dispersion from the IBES/Compustat sample. I then use the calibrated model to investigate how much capital–labor-ratio and marginal-product dispersion communication friction generates at the estimated parameters, and how the resulting distortions affect aggregate TFP.

The quantitative analysis highlights a distinction between communication friction and common information or behavioral frictions. When managers share the same noisy information and behavioral expectations, their input choices respond to a common productivity belief. In the absence of other frictions, this common belief changes the scale of inputs without altering the capital–labor ratio, much like a revenue distortion in Hsieh and Klenow (2009). Communication friction instead creates differences between the beliefs underlying capital and labor decisions. Investment and hiring need not move together, and the capital–labor ratio varies with the managers' relative beliefs. In this sense, communication friction is akin to an input-specific distortion that changes the capital-labor ratio of the firm.

This distinction is quantitatively important. In the calibrated model, managers place substantially more weight on their private signals than on the current common signal. Their productivity beliefs therefore need not move together, weakening the comovement of investment and hiring. In a counterfactual retaining only communication friction at its estimated parameters, the investment–hiring correlation is 0.33, much lower than other mechanisms would generate by themselves. Moreover, communication friction appears to be an important driver of capital-labor ratio variability: it generates approximately 39 percent of the observed dispersion in capital-labor ratio by itself. By comparison, common noisy information combined with over-extrapolative beliefs generates perfectly correlated investment and hiring and no capital–labor-ratio dispersion when operating alone. These results show that communication friction provides a quantitatively relevant source of input-specific distortions. Restricting imperfect information to a common firm-level belief excludes this channel and risks attributing the resulting input-mix variation to other mechanisms, such as financial frictions or non-neutral technologies.

Communication friction also generates sizable marginal-product dispersion and aggregate productivity loss. When other frictions are muted, communication friction generates approximately 27 percent of observed MRPK dispersion, which is substantially higher than the MRPK dispersion generated by convex costs and correlated distortions. It generates about 19 percent of the observed MRPN dispersion, which is similar in magnitude to that generated by convex adjustment costs. The associated log TFP loss of communication friction is about 5 percent relative to the frictionless allocation.

The negative covariance between MRPK and MRPN limits this productivity loss. Communication friction generates substantially more MRPK dispersion than convex adjustment costs alone, yet its associated TFP loss is smaller. This is because communication friction generates a strongly negative MRPK–MRPN correlation, whereas convex adjustment costs and common information generate positive correlations. This negative correlation between marginal revenue products arises from managers' reliance on private information under communication friction. Holding actual productivity and the labor manager's information fixed, a favorable realization of the capital manager's private-signal noise raises capital investment. The additional capital lowers MRPK through diminishing returns but raises MRPN because capital and labor are complements, resulting in a negative correlation in marignal revenue products. This finding echoes the point made by David and Venkateswaran (2019) that assessing the aggregate consequences of distortions requires examining not only the dispersion but also the covariance of marginal revenue products they generate.

Related Literature This paper is closely related to three strands of literature. First, it contributes to the literature on management and organizational economics. Existing works (e.g. Bloom, Brynjolfsson, Foster, Jarmin, Patnaik, Saporta-Eksten, and Van Reenen (2019), Guner, Parkhomenko, and Ventura (2018), Sadun, Schuh, Hartley, Reenen, and Bloom (2025)) focus on how broad measures of management practices correlate with firm productivity.[2] My paper documents a particular channel using firm expectations. Existing empirical work studies within-firm communication using geographic variation in access to information and detailed communication records, as in Giroud (2013) and Impink, Prat, and Sadun (2024). I develop a complementary approach that uses the predictability of investment forecast error to test whether information is shared across decision-makers. The approach requires neither access to internal communication records nor forecasts from multiple managers within each firm, making it applicable to broader firm panels with the expectation and outcome data. Embedding this test in a quantitative model connects evidence on firms' internal information structure to investment, hiring, and aggregate productivity.

Second, the paper contributes to the literature on information frictions and behavioral expectations in macroeconomics. Empirically, this paper's strategy of identifying communication frictions is built upon forecast-error tests proposed by Coibion and Gorodnichenko (2012, 2015) and Bordalo, Gennaioli, Ma, and Shleifer (2020), which focus on the macroeconomic expectation of firms and professional forecasters. Kohlhas and Walther (2021) highlights the importance of considering the forecast error predictability by different sources jointly. Build-

[2] For instance, the management score in Bloom et al. (2019) is the composite of 16 questions on structured management, ranging from incentives, performance, decentralization and data use.

ing on these approaches, I derive joint restrictions on own- and cross-decision forecast-error predictability that distinguish segmented information from a common-information benchmark with behavioral belief distortions. Theoretically, the dynamic team production framework is developed from Marschak and Radner (1972) and Lian (2020), and applies the insights from the macroeconomic literature of imperfect coordination (e.g. Cooper and John (1988), Woodford (2001), Angeletos and Lian (2016)) to investigate within-firm frictions. The framework resolves the puzzle I document in the test: investment forecasts simultaneously overreact to investment and underreact to hiring. Quantitatively, the paper complements Ma, Ropele, Sraer, and Thesmar (2020) and Barrero (2022) which structurally estimate information/behavioral frictions by targeting expectation moments.

Lastly, the paper contributes to the literature on the implications of information and behavioral frictions for allocative efficiency. David, Hopenhayn, and Venkateswaran (2016) and David and Venkateswaran (2019) quantify the contribution of information friction to misallocation under two types of internal information structure: either there is a firm-level noisy signal that is commonly known by the managers, or the labor decision is made ex post and only capital decisions are subject to information frictions. My paper generalizes the analysis and can accommodate any given information structure between capital and labor managers within the firm. Recently, Gautier et al. (2025) and Ropele et al. (2024) considers the misallocation and TFP loss resulting from sales and inflation expectations, in which information frictions affect capital and labor symmetrically and do not distort the capital-labor ratio by themselves. The mechanism in my paper highlights that communication friction is an input-specific distortion that generates large variations in capital-labor ratio.

Layout The remainder of the paper is organized as follows. Section 2 presents the theory. Section 3 describes the empirical test for within-firm communication friction and applies it to U.S. public firms. Section 4 quantifies the aggregate implications of communication frictions, discussing parameter identification and calibration results. Section 5 concludes.

2 Model

In this section, I describe the model of dynamic input choices under communication friction. Throughout the paper, there is no aggregate uncertainty, so that all the aggregate variables are fixed at their steady-state values in equilibrium and are common knowledge to all the agents in the economy.

Household There is a representative household in the economy, who has standard preference on final-good consumption, discounts future consumption utility by a discount factor β and supplies a fixed amount of labor $\bar{N}$ to the firms at a wage rate W_t. He can save in a risk-free bond at a real interest rate r in each period. The household chooses how much to save and consume subject to a budget constraint. In a stationary equilibrium, consumption is constant over time, and by the Euler equation, we have $\beta(1+r) = 1$. The only role of the household sector is to close the model, and it is not essential to the analysis.

Final-Good Producer There is a final good producer who combines intermediate goods $\{Y_{it}\}$ into final good Y_t by a CES technology

$$Y_t = \left(\int_0^1 (A_{it} Y_{it})^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}} \tag{1}$$

where A_{it} can be interpreted as an idiosyncratic productivity shock on intermediate-good producer i or a demand shock on the product i, and $\epsilon > 1$ is the parameter for elasticity of substitution.

The price of final good is normalized to one. Given price $\{P_{it}\}$, the final good producers choose how much intermediate goods to purchase from intermediate-good producers. This yields a standard demand curve

$$P_{it} = \left(\frac{Y_t}{Y_{it}} \right)^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} \tag{2}$$

Intermediate-Good Producer There is a continuum $i \in [0, 1]$ of intermediate-good producers in the economy. Each producer i has a constant-return-to-scale production function to combine capital and labor inputs into intermediate goods:

$$Y_{it} = K_{it}^{\alpha} N_{it}^{1-\alpha} \tag{3}$$

where $\alpha \in (0, 1)$ is the parameter for capital share. Each producer i operates in a standard monopolistic competition environment: it internalizes the demand function (2), so that its revenue can be expressed as a decreasing-return-to-scale function of capital and labor, denoted as $R(K_{it}, N_{it}; A_{it})$:

$$R(K_{it}, N_{it}; A_{it}) = P_{it}Y_{it} = Y_t^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} Y_{it}^{1-\frac{1}{\epsilon}} = Y_t^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} K_{it}^{\hat{\alpha}_1} N_{it}^{\hat{\alpha}_2} \tag{4}$$

where $\hat{\alpha}_1 \equiv \alpha(1 - 1/\epsilon)$ and $\hat{\alpha}_2 \equiv (1 - \alpha)(1 - 1/\epsilon)$.

Producer i's cost of production includes two parts. First, the firm pays the rental cost of capital and hiring cost of labor. Additionally, the firm faces convex adjustment costs

$$\frac{\xi_k}{2}\left(\frac{I_{it}}{K_{i,t-1}} - \delta\right)^2 K_{i,t-1} \text{ and } \frac{\xi_n}{2}\left(\frac{H_{it}}{N_{i,t-1}}\right)^2 N_{i,t-1}$$

where $I_{it} \equiv K_{it} - (1 - \delta)K_{i,t-1}$ is investment and $H_{it} \equiv N_{it} - N_{i,t-1}$ is hiring. ξ_k and ξ_n are parameters that control the slope of marginal adjustment cost for capital and labor respectively. Putting everything together, the cost function $\mathcal{C}(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1})$ is

$$\mathcal{C}(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1}) = R_tK_{it} + W_tN_{it} + \frac{\xi_k}{2}\left(\frac{I_{it}}{K_{i,t-1}} - \delta\right)^2 K_{i,t-1} + \frac{\xi_n}{2}\left(\frac{H_{it}}{N_{i,t-1}}\right)^2 N_{i,t-1}$$

where δ is the deprecation rate, and R_t is the rental rate of capital, which is equal to $r + \delta$ in a stationary equilibrium.

Denote profit function $\Pi(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1}, A_{it}) \equiv R(K_{it}, N_{it}; A_{it}) - \mathcal{C}(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1})$. Under full information, the firm's input choices are characterized by the Bellman equation

$$V(K_{i,t-1}, N_{i,t-1}; A_{it}) = \max_{K_{it}, N_{it}} \Pi(K_{it}, N_{it}; K_{i,t-1}, N_{i,t-1}, A_{it}) + \frac{1}{1+r}\mathbb{E}[V(K_{it}, N_{it}; A_{i,t+1})|A_{it}] \tag{5}$$

To facilitate my later discussions, it is useful to think of the firm as a team of two managers, one in charge of capital decisions (I call him the Capital Manager) and the other in charge of labor decisions (I call her the Labor Manager), who share the common objective of maximizing the firm's lifetime profit. Hence, the capital and labor policy function that solves the *firm's* Bellman equation (5) are equivalent to a Markov Perfect Equilibrium (MPE) of a dynamic team production game (Marschak and Radner (1972), Chapter 7), in which

- Given the labor manager's policy N_{it}, the capital manager's strategy K_{it} is the one that solves the Bellman equation

$$V^k(K_{i,t-1}, N_{i,t-1}; A_{it}) = \max_K \Pi(K, N_{it}; K_{i,t-1}, N_{i,t-1}, A_{it}) + \frac{1}{1+r}\mathbb{E}[V^k(K, N_{it}; A_{i,t+1})|A_{it}] \tag{6}$$

- Given the capital manager's policy K_{it}, the capital manager's strategy N_{it} is the one that solves the Bellman equation

$$V^n(K_{i,t-1}, N_{i,t-1}; A_{it}) = \max_N \Pi(K_{it}, N; K_{i,t-1}, N_{i,t-1}, A_{it}) + \frac{1}{1+r}\mathbb{E}[V^n(K_{it}, N; A_{i,t+1})|A_{it}] \tag{7}$$

Equation (6) and (7) are the *manager*-level Bellman equations that characterize the firm's input decisions. This reformulation gives us more flexibility to accommodate incomplete/asymmetric information between managers within the firm and makes it easier for us to analyze a richer internal information structure within the firm.

Information The only source of uncertainty in this simple model is the idiosyncratic firm fundamentals A_{it}. I assume that the log of A_{it}, denoted as a_{it}, follows an AR(1) process

$$a_{it} = \rho a_{i,t-1} + \mu_{it}, \text{ with } \mu_{it} \overset{iid}{\sim} N(0, \sigma_\mu^2) \tag{8}$$

The firm managers know the data generating process in (8) and treat the objective distribution of μ_{it} or a_{it} as the prior. However, the true value of a_{it} is never fully revealed to the firm and its managers.[3] Hence, when making decisions, the capital and labor manager of firm i need to learn dynamically from the Gaussian noisy signals they receive to form expectations about the "latent variable" a_{it} over time using Kalman filters, which I will specify later in detail.

The key friction is that I allow each manager's decision to be dependent on different, non-nested information sets. Formally, let $\mathcal{I}_{it}^k$ denote capital manager's information set in period t, and $\mathcal{I}_{it}^n$ denote the labor manager's information set in period t. In particular, I allow the possibility that $\mathcal{I}_{it}^k \neq \mathcal{I}_{it}^n$ and are not a subset to each other. This captures the notion of imperfect communication within the organization: with imperfect information sharing, the managers' information sets do not perfectly overlap.

[3] This is different from the setup in David, Hopenhayn, and Venkateswaran (2016) and David and Venkateswaran (2019), where the lagged firm fundamental $a_{i,t-1}$ is fully revealed to the firm in period t.

I now specify the elements in the managers' information sets $\mathcal{I}_{it}^k$ and $\mathcal{I}_{it}^n$. I assume that there is a piece of firm-level information s_{it}^p, which is a noisy signal of the firm's fundamental a_{it} and is commonly known by the managers (i.e. $s_{it}^p \in \mathcal{I}_{it}^k \cap \mathcal{I}_{it}^n$):

$$s_{it}^p = a_{it} + \epsilon_{it}^p, \text{ with } \epsilon_{it}^p \overset{iid}{\sim} N(0, \sigma_{\epsilon,p}^2) \tag{9}$$

For the non-nested part of the information set, I assume that each manager $m \in \{k, n\}$ is endowed with a private, noisy signal s_{it}^m about a_{it}:

$$s_{it}^m = a_{it} + \epsilon_{it}^m, \text{ with } \epsilon_{it}^m \overset{iid}{\sim} N(0, \sigma_{\epsilon,m}^2) \text{ for } m \in \{k, n\}. \tag{10}$$

The signal noises $\epsilon_{it}^k, \epsilon_{it}^n, \epsilon_{it}^p$ and the innovation in the firm fundamental μ_{it} are independent to each other. In each period, the realization of s_{it}^k and s_{it}^n are private to the capital and labor manager respectively, but the data generating process of the signals, (9) and (10), are commonly known by the managers.

The standard deviations of the signal noises, $(\sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n})$, define the internal information structure of the firm and are the key parameters to pin down in the quantitative analysis. This parametrization gives us a parsimonious way to model the degree of information segmentation within the firm. For example, if we make the private signals to be extremely noisy, i.e. $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$, then the managers will put essentially zero weight on private signals when they apply Bayes's rule to update their beliefs, and the firm's behavior is as if the firm is endowed with a public information only and the managers' information set perfectly overlaps. On the other hand, if $\sigma_{\epsilon,p} \to \infty$ while $\sigma_{\epsilon,m} > 0$, then the managers will not rely on the firm-level signal to update their beliefs, and it's as if the firm is operating on a non-nested information structure.

Since the realization of signal s_{it}^m is private to manager m and both managers choose their decisions simultaneously, the action that manager m takes in period t is not known by the other manager m', and vice versa. However, I assume that at the end of each period t, the actions taken by the managers in period t, K_{it} and N_{it}, are revealed to both managers and become public information. This assumption allows the managers to use the observed actions in the previous period to update their beliefs about the unknown fundamental a_{it} and the other manager's action in the current period, which simplifies the characterization of the equilibrium concept. Formally, putting everything together, we can express the law of motion for the managers' information sets $\mathcal{I}_{it}^k$ and $\mathcal{I}_{it}^n$ as

$$\mathcal{I}_{it}^k = \mathcal{I}_{i,t-1}^k \cup \{s_{it}^p, s_{it}^k, N_{i,t-1}\} \text{ and } \mathcal{I}_{it}^n = \mathcal{I}_{i,t-1}^n \cup \{s_{it}^p, s_{it}^n, K_{i,t-1}\} \tag{11}$$

Equilibrium Under communication friction, the firm faces a dynamic team production problem with incomplete information. The firm's input decisions can therefore be interpreted as a Markov Perfect Bayesian Equilibrium (MPBE), in which

- Given labor manager's policy function $N_{it}(\cdot)$, capital manager's policy function $K_{it} : \mathcal{I}^k_{it} \to \mathbb{R}^+$ solves Bellman equation

  $$V^k(K_{i,t-1}, N_{i,t-1}; \mathcal{I}^k_{it}) = \max_K \mathbb{E}\left[\Pi(K, N_{it}(s^n_{it}); K_{i,t-1}, N_{i,t-1}, A_{it}) + \beta V^k(K, N_{it}(s^n_{it}); \mathcal{I}^k_{i,t+1}) \middle| \mathcal{I}^k_{it}\right] \tag{12}$$

  where s^n_{it} is the signal realization of the labor manager in period t, and

- Given capital manager's policy function $K_{it}(\cdot)$, labor manager's policy function $N_{it} : \mathcal{I}^n_{it} \to \mathbb{R}^+$ solves Bellman equation

  $$V^n(K_{i,t-1}, N_{i,t-1}; \mathcal{I}^n_{it}) = \max_N E_t\left[\Pi(K_{it}(s^k_{it}), N; K_{i,t-1}, N_{i,t-1}, A_{it}) + \beta V^n(K_{it}(s^k_{it}), N; \mathcal{I}^n_{i,t+1}) \middle| \mathcal{I}^n_{it}\right] \tag{13}$$

  where s^k_{it} is the signal realization of the capital manager in period t, and

- In period t, each manager updates his/her belief about a_{it} using his/her observed signals and the past actions $K_{i,t-1}, N_{i,t-1}$, consistent with the policy function $K_{it}(\cdot), N_{it}(\cdot)$ and the Bayes's rule.

The advantage of using MPBE in (12) and (13) to characterize firm's behavior lies in its flexibility of accommodating essentially any possible internal information structure of the firm. This flexibility makes it a more general model and nests two classes of extensively studied models in the firm dynamics literature. One is the models based on full-information rational expectation (FIRE), as in (5) or (6) and (7) above. They can be regarded as a special case of (12) and (13), with $\mathcal{I}^k_{it} = \mathcal{I}^n_{it} = \{a_{it}, a_{i,t-1}, \cdots, a_{i0}\}$, i.e. the true firm fundamental is always revealed to both managers. Using our parameterization (9) and (10) above, this full-information benchmark maps to the limit $\sigma_{\epsilon,p} \to 0$ and $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$. Another class of models are the ones based on common, noisy information, as in David, Hopenhayn, and Venkateswaran (2016).[4] Although it is a useful benchmark to analyze the effect of uncertainty and imperfect information on firm dynamics, it presumes

[4]More precisely, David et al. (2016) studies two extreme cases, where (1) imperfectly observed information is perfectly shared among capital and labor managers, and (2) labor decisions are made *ex post* and frictionlessly, and only capital decisions are subject to information friction (which is also the main specification of David and Venkateswaran (2019)). The latter case is also a special case of the current framework, with the capital manager's information set is a subset of the labor manager's information set. My paper can be regarded as providing a smooth version of information friction between these two extreme cases.

that that imperfectly observed information is frictionlessly shared and communicated within the organization. From the above discussion, this is again a special case of the current framework, with $\mathcal{I}^k_{it} = \mathcal{I}^n_{it} = \{s^p_{it}, \cdots, a^p_{i0}\}$. Using our parameterization (9) and (10) above, this noisy, common information benchmark maps to the limit $\sigma_{\epsilon,p} \in (0, \infty)$ and $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$. Hence, the firm dynamics implied by noisy, common information models is also contained in (12) and (13).

Equation (12) and (13) illustrate that the managers face two types of uncertainty when they are making their own decisions. For the capital manager, apart from the fundamental uncertainty about a_{it} or A_{it}, he faces a strategic uncertainty about the action of the labor manager N_{it}, which is measurable to labor manager's information set only. The source of this strategic uncertainty is the fact that information is not perfectly shared within the firm, so that there is labor-specific information that is only known to the labor manager and not to the capital manager. This strategic uncertainty affects capital manager's decision because labor manager's action affects both the marginal revenue product of capital and the continuation value of the capital manager, and therefore capital manager has an incentive to form expectation about N_{it} to guide his own decision about K_{it}.

Log-Quadratic Approximation The third requirement of an MPBE requires that the belief dynamics of both managers are consistent with the policy function and Bayes's rule. It is a difficult task to analytically characterize or compute the evolution of beliefs since it is an infinite-dimensional object.

Here, I get around this technical challenge by doing a Woodford (2003)-type log-quadratic approximation on the profit function with respect to $(K_{it}, N_{it}, K_{i,t-1}, N_{i,t-1}, A_{it})$ around their frictionless steady-state values.[5] I can therefore express the log-quadratic approximated profit function π_{it} as a quadratic function in terms of the log deviations of capital, labor and firm fundamental from their own frictionless steady-state value. Denote these log-deviation values as k_{it}, n_{it} and a_{it} respectively, and define investment (rate) as $\iota_{it} \equiv k_{it} - k_{i,t-1}$ and hiring (rate) as $h_{it} \equiv n_{it} - n_{i,t-1}$. The approximated profit function can be written as

$$\pi(\iota_{it}, h_{it}; x_{it}) = x'_{it} P x_{it} + x'_{it} Q \iota_{it} + x'_{it} R h_{it} + H_\iota \iota^2_{it} + H_{\iota h} \iota_{it} h_{it} + H_h h^2_{it} \tag{14}$$

[5] By frictionless, I mean the case where there is no information friction and adjustment friction, i.e. a_{it} is fully revealed to the firm, and $\xi_k = \xi_n = 0$.

where $x_{it} = [k_{i,t-1}, n_{i,t-1}, a_{it}]'$ is the state vector of the firm with a law of motion

$$x_{i,t+1} = Ax_{it} + B\iota_{it} + Ch_{it} + D\mu_{i,t+1} \tag{15}$$

and matrices $A, B, C, D, P, Q, R, H_\iota, H_h$ and $H_{\iota h}$ are constants imputed from the frictionless steady-state values and model parameters. The derivations of (14) are left to Appendix A.1.

Characterization of Linear MPBE The log-quadratic approximation of profit function allows us to characterize a linear MPBE analytically.

Plugging (14) and (15) back into the manager-level Bellman equations (12) and (13) transforms the original problem into a linear-quadratic (LQ) control problem, with investment ι_{it} and hiring h_{it} as control variables and x_{it} as the state vector. An immediate implication of the LQ control is that the policy function is linear and certainty equivalence holds:

Lemma 1. *With the log-quadratic-approximated profit function (14) and the law of motion for state (15), we have*

1. *The policy function for the full-information MPE in (6) and (7) takes the form*

$$\begin{aligned} \iota_{it} &= F_\iota x_{it} = F_\iota^k k_{i,t-1} + F_\iota^n n_{i,t-1} + F_\iota^a a_{it} \\ h_{it} &= F_h x_{it} = F_h^k k_{i,t-1} + F_h^n n_{i,t-1} + F_h^a a_{it} \end{aligned}$$

   where the policy matrix F_ι, F_h can be solved by iterating forward a pair of Ricatti equations;

2. *The policy function for (12) and (13) under communication friction takes the form*

$$\iota_{it} = F_\iota \mathbb{E}[x_{it}|\mathcal{I}_{it}^k] = F_\iota^k k_{i,t-1} + F_\iota^n n_{i,t-1} + F_\iota^a \hat{a}_{it}^k \tag{16}$$

$$h_{it} = F_h \mathbb{E}[x_{it}|\mathcal{I}_{it}^n] = F_h^k k_{i,t-1} + F_h^n n_{i,t-1} + F_h^a \hat{a}_{it}^n \tag{17}$$

   where F_ι, F_h are the same policy matrix as in the full-information MPE, and $\hat{a}_{it}^k \equiv \mathbb{E}[a_{it}|\mathcal{I}_{it}^k]$ and $\hat{a}_{it}^n \equiv \mathbb{E}[a_{it}|\mathcal{I}_{it}^n]$ are the posterior mean of capital and labor manager's first-order belief on a_{it}.

Proof. See Appendix A.3. □

The linearity of (16) and (17), together with the Gaussian structure of μ_{it} and signal noises, significantly simplifies the characterization of the belief updating process for two reasons. First, the Gaussian shocks and noises imply that tracking the first and second moment is sufficient for the characterization of belief dynamics, which significantly lowers the dimensions needed for the computation. Moreover, the linearity of policy functions makes actions jointly Gaussian with firm fundamentals and signals, and the managers can simply treat the other manager's lagged action as a noisy signal of the firm fundamentals and learn from it in the same manner as from other signals. For instance, in period t, with (17), the capital manager forms belief about the labor manager's action h_{it} by

$$\hat{h}_{it}^{k} \equiv \mathbb{E}[h_{it}|\mathcal{I}_{it}^{k}] = F_{h}^{k} k_{i,t-1} + F_{h}^{n} n_{i,t-1} + F_{h}^{a} \mathbb{E}[\hat{a}_{it}^{n}|\mathcal{I}_{it}^{k}] \tag{18}$$

Equation (18) conveys two important messages. First, the capital manager's forecast of hiring depends on his *second-order belief* about the labor manager's nowcast of a_{it}, $\hat{a}_{it}^{(k,n)} \equiv \mathbb{E}[\hat{a}_{it}^{n} \mid \mathcal{I}_{it}^{k}]$. Second, the loading F_{h}^{a} determines whether observed hiring reveals the labor manager's posterior mean. Under the scalar Gaussian structure below, a nonzero loading makes the posterior mean, and hence the underlying private signal, invertible from the action. Thus second-order beliefs are needed when decisions are made, but they can be computed directly from first-order posteriors and need not be carried as a recursive covariance state. The same argument applies to the labor manager's belief about the capital manager's investment decision.

The scalar structure of the latent state further simplifies the belief dynamics. Suppose that $F_{\iota}^{a} \neq 0$ and $F_{h}^{a} \neq 0$. Because predetermined inputs are publicly observed at the beginning of each period, the linear decision rules (16) and (17) imply that observing a manager's action reveals that manager's posterior mean about a_{it}. Moreover, because the posterior mean is an invertible function of the manager's private signal, the action also reveals that private signal. Consequently, after the period-$t-1$ actions become public, both managers possess the public signal and both private signals from period $t-1$ and therefore share a common prior about a_{it}. The following lemma characterizes the evolution of beliefs.

Lemma 2. *Suppose that the latent state a_{it} is scalar and the linear policy coefficients satisfy $F_{\iota}^{a} F_{h}^{a} \neq 0$. Let $\bar{\Sigma}$ be the fixed point of*

$$\bar{\Sigma} = \left[\left(\rho^{2}\bar{\Sigma} + \sigma_{\mu}^{2}\right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2} + \sigma_{\epsilon,n}^{-2}\right]^{-1}$$

and $\hat{\Sigma}^{k} \equiv \left((\rho^{2}\bar{\Sigma} + \sigma_{\mu}^{2})^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2}\right)^{-1}$, $\hat{\Sigma}^{n} \equiv \left((\rho^{2}\bar{\Sigma} + \sigma_{\mu}^{2})^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,n}^{-2}\right)^{-1}$. *The man-*

agers' posterior beliefs are

$$\hat{a}_{it}^{k} = \hat{\Sigma}^{k}\left((\rho^2\bar{\Sigma}+\sigma_{\mu}^2)^{-1}\bar{a}_{it}+\sigma_{\epsilon,p}^{-2}s_{it}^{p}+\sigma_{\epsilon,k}^{-2}s_{it}^{k}\right), \tag{19}$$

$$\hat{a}_{it}^{n} = \hat{\Sigma}^{n}\left((\rho^2\bar{\Sigma}+\sigma_{\mu}^2)^{-1}\bar{a}_{it}+\sigma_{\epsilon,p}^{-2}s_{it}^{p}+\sigma_{\epsilon,n}^{-2}s_{it}^{n}\right), \tag{20}$$

and their second-order beliefs are

$$\hat{a}_{it}^{(k,n)} = \hat{\Sigma}^{n}\left((\rho^2\bar{\Sigma}+\sigma_{\mu}^2)^{-1}\bar{a}_{it}+\sigma_{\epsilon,p}^{-2}s_{it}^{p}+\sigma_{\epsilon,n}^{-2}\hat{a}_{it}^{k}\right), \tag{21}$$

$$\hat{a}_{it}^{(n,k)} = \hat{\Sigma}^{k}\left((\rho^2\bar{\Sigma}+\sigma_{\mu}^2)^{-1}\bar{a}_{it}+\sigma_{\epsilon,p}^{-2}s_{it}^{p}+\sigma_{\epsilon,k}^{-2}\hat{a}_{it}^{n}\right) \tag{22}$$

The corresponding conditional variances are

$$\hat{\Sigma}^{(k,n)} = \left(\hat{\Sigma}^{n}\sigma_{\epsilon,n}^{-2}\right)^2\left(\hat{\Sigma}^{k}+\sigma_{\epsilon,n}^{2}\right),$$

$$\hat{\Sigma}^{(n,k)} = \left(\hat{\Sigma}^{k}\sigma_{\epsilon,k}^{-2}\right)^2\left(\hat{\Sigma}^{n}+\sigma_{\epsilon,k}^{2}\right).$$

Here $\bar{a}_{it}$, the common prior of the period, evolves according to

$$\bar{a}_{it} = \rho\bar{\Sigma}\left((\rho^2\bar{\Sigma}+\sigma_{\mu}^2)^{-1}\bar{a}_{i,t-1}+\sigma_{\epsilon,p}^{-2}s_{i,t-1}^{p}+\sigma_{\epsilon,k}^{-2}s_{i,t-1}^{k}+\sigma_{\epsilon,n}^{-2}s_{i,t-1}^{n}\right). \tag{23}$$

Proof. See Appendix A.2. □

Now we have all the ingredients to characterize a linear MPBE associated with (14) and (15).

Lemma 3. *Suppose that $F_\iota^a F_h^a \neq 0$. A linear Markov Perfect Bayesian Equilibrium is a set of policies (F_ι, F_h) and Gaussian beliefs summarized by $(\hat{a}_{it}^{k}, \hat{\Sigma}^{k})$, $(\hat{a}_{it}^{n}, \hat{\Sigma}^{n})$, $(\hat{a}_{it}^{(k,n)}, \hat{\Sigma}^{(k,n)})$, and $(\hat{a}_{it}^{(n,k)}, \hat{\Sigma}^{(n,k)})$ such that*

- *Beliefs are consistent with strategies: given (F_ι, F_h), the managers update their first- and second-order beliefs by Bayes' rule according to Lemma 2;*
- *Strategies are consistent with beliefs: given these beliefs, (F_ι, F_h) solve the managers' Bellman equations, and investment and hiring satisfy (16) and (17), as in Lemma 1.*

Proof. See Appendix A.4. □

The linear MPBE in Lemma 3 implies a law of motion for the joint distribution of action, expectation and state, namely, the vector $\equiv [k_{it}, n_{it}, \hat{a}^k_{it}, \hat{a}^n_{it}, a_{it}]'$. A stationary equilibrium requires this distribution to be at its stationary distribution. The computation of this stationary distribution is essential to the quantitative analysis in Section 4 and the details of computation are left to Appendix A.5.

3 Identifying Communication Friction

In this section, I describe the empirical strategy of identifying communication friction from managerial expectation data and implement the methodology to U.S. public firms.

3.1 Predictability of Investment Forecast Error

The key object that maps the theory to data is the investment forecast error, defined as the difference between actual capital investment ι_{it} and the capital manager's one-period-ahead forecast of capital investment, $\mathbb{F}^k_{i,t-1}[\iota_{it}]$. Note that the expectation operator $\mathbb{F}^k_{i,t-1}$ is allowed to deviate from the rational expectation $\mathbb{E}^k_{i,t-1} \equiv \mathbb{E}[\cdot|\mathcal{I}^k_{i,t-1}]$. The investment forecast error can be decomposed as

$$\iota_{it} - \mathbb{F}^k_{i,t-1}[\iota_{it}] = \underbrace{\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}]}_{\text{Rational Forecast Error}} + \underbrace{\mathbb{E}^k_{i,t-1}[\iota_{it}] - \mathbb{F}^k_{i,t-1}[\iota_{it}]}_{\text{Behavioral Distortion}}. \tag{24}$$

The rational forecast error is the difference between actual investment and rational forecast of investment. It captures the contribution of information friction to the forecast error. The behavioral distortion term is the wedge between rational and behavioral forecast. It captures the contribution of deviations from rational expectation to the forecast error.

The behavioral macro literature (e.g. Coibion and Gorodnichenko (2012, 2015)) establishes that the predictability of forecast errors by past variables is informative of the types of behavioral and informational frictions in the expectation. Here, I consider two regressions. The first one asks whether the investment forecast error can be predicted by past investment:

$$\iota_{it} - \mathbb{F}^k_{i,t-1}[\iota_{it}] = \alpha_\iota + \beta_\iota \iota_{i,t-1} + \nu_{it} \tag{25}$$

and the second one considers whether the investment forecast error can be predicted by past hiring:

$$\iota_{it} - \mathbb{F}^k_{i,t-1}[\iota_{it}] = \alpha_h + \beta_h h_{i,t-1} + \nu_{it} \tag{26}$$

The coefficients of interest are β_ι and β_h. Using (24), I decompose these regression coefficients as

$$\beta_\iota = \underbrace{\frac{\mathrm{cov}(\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}], \iota_{i,t-1})}{\mathrm{var}(\iota_{i,t-1})}}_{\equiv \beta^R_\iota} + \underbrace{\frac{\mathrm{cov}(\mathbb{E}^k_{i,t-1}[\iota_{it}] - \mathbb{F}^k_{i,t-1}[\iota_{it}], \iota_{i,t-1})}{\mathrm{var}(\iota_{i,t-1})}}_{\equiv \beta^B_\iota} \tag{27}$$

and

$$\beta_h = \underbrace{\frac{\mathrm{cov}(\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}], h_{i,t-1})}{\mathrm{var}(h_{i,t-1})}}_{\equiv \beta^R_h} + \underbrace{\frac{\mathrm{cov}(\mathbb{E}^k_{i,t-1}[\iota_{it}] - \mathbb{F}^k_{i,t-1}[\iota_{it}], h_{i,t-1})}{\mathrm{var}(h_{i,t-1})}}_{\equiv \beta^B_h} \tag{28}$$

Here, β^R_ι and β^R_h capture the predictability of investment forecast error resulting from the predictability of rational forecast error, and β^B_ι and β^B_h capture the predictability of investment forecast error resulting from behavioral distortions.

Own-decision Predictability I first analyze the own-decision predictability β_ι. Since the capital manager's investment $\iota_{i,t-1}$ in period $t-1$ is based on his information set $\mathcal{I}^k_{i,t-1}$ in that period, it is included in the information set in $t-1$. By the Law of Iterated Expectation, we have $\mathrm{cov}(\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}], \iota_{i,t-1}) = 0$ and hence $\beta^R_\iota = 0$. In other words, the own-decision predictability β_ι comes solely from the deviations from rational expectation. Specifically, a negative β_ι means that the reported forecast $\mathbb{F}^k_{i,t-1}$ responds more strongly to the news that leads to high investment $\iota_{i,t-1}$ than the rational forecast does. This is often referred to as "belief overreaction" in the behavioral macro literature and is in favor of mechanisms such as diagnostic expectation (e.g. Bordalo et al. (2020)) and over-extrapolation (e.g. Angeletos et al. (2021), Barrero (2022)). As I will show later, this is the empirically relevant case for the U.S. public firms. On the other hand, a positive β_ι means that the reported forecast $\mathbb{F}^k_{i,t-1}$ responds less strongly to high investment $\iota_{i,t-1}$ than the rational forecast, which is in favor of mechanisms such as under-extrapolation and cognitive discounting (e.g. Gabaix (2019)).

Cross-decision Predictability Unlike own-decision predictability β_ι, the cross-decision predictability β_h reflects not only behavioral expectations but also communication frictions within the firm. The communication friction affects β_h via the rational forecast error channel β^R_h. From the linear investment decision rule (16) and the belief updating equation (19) for the capital manager, we have the following proposition that characterizes the structure of investment forecast error:

Proposition 1. *In a MPBE defined in Lemma 3, the rational forecast error of firm i's investment takes the form*

$$\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}] = \beta_1(a_{i,t-1} - \mathbb{E}^k_{i,t-1}[a_{i,t-1}]) + \beta_2(h_{i,t-1} - \mathbb{E}^k_{i,t-1}[h_{i,t-1}]) + \beta_3(H\mu_{it} + w^k_{it}) \quad (29)$$

where $\beta_1, \beta_2 \in \mathbb{R}$ and $\beta_3 \in \mathbb{R}^{1\times 2}$ are constants determined by the policy coefficients in Lemma 1 and the posterior variances in Lemma 2. Their explicit expressions are given in Appendix A.6.

Proof. See Appendix A.6. □

Proposition 1 shows that the rational investment forecast error has three components. The first term captures the fundamental uncertainty that the capital manager faces. It is the gap between actual productivity $a_{i,t-1}$ and the capital manager's expectation in period $t-1$ about the productivity in that period. As long as the capital manager is uncertain about the true productivity in period $t-1$, this gap will not be zero. The third term $\beta_3(H\mu_{it} + w^k_{it})$ collects all the time-t productivity innovations and signal noise, which are orthogonal to information on or before period $t-1$.

The second term, $\beta_2(h_{i,t-1} - \mathbb{E}^k_{i,t-1}[h_{i,t-1}])$, speaks directly to the communication friction. It is the gap between actual hiring in period $t-1$ and the capital manager's expectation in period $t-1$ about $h_{i,t-1}$. Absent communication friction, this term will always equal zero. This is because the capital manager knows that the labor manager's decision is based on the same information as his, and therefore he knows the labor manager's decision in $t-1$. His expectation in $t-1$ about hiring in that period, $\mathbb{E}^k_{i,t-1}[h_{i,t-1}]$, will therefore be exactly the same as $h_{i,t-1}$. On the other hand, if communication is imperfect within the organization, the capital manager does not observe the exact $h_{i,t-1}$ in period $t-1$ because of the labor manager's private information. As a result, his belief about the labor manager's action in that period generally differs from the labor manager's actual decision, making this strategic uncertainty contribute to the investment forecast error.

Let's now consider the implications of Proposition 1 for the cross-decision regression coefficient β_h. Under common information, previous period's hiring $h_{i,t-1}$ is included in the capital manager's information set $\mathcal{I}^k_{i,t-1}$. As a result, we have $\text{cov}(a_{i,t-1} - \mathbb{E}^k_{i,t-1}[a_{i,t-1}], h_{i,t-1}) = 0$ by the Law of Iterated Expectation and $h_{i,t-1} = \mathbb{E}^k_{i,t-1}[h_{i,t-1}]$. Therefore, the rational forecast error of investment should not be predicted by past hiring (i.e. $\beta_h^R = 0$). Hence, under common information in the baseline model with a scalar Hicks-neutral fundamental, both own-decision and cross-decision predictability are driven by the same behavioral distortion. Because investment and hiring respond in the same direction to productivity beliefs, the common behavioral distortion tends to generate the same sign for β_ι and β_h. To formalize this point, consider the case where both managers receive only the common, noisy signal s_{it}^p in each period and use the common behavioral belief $\mathbb{F}_{it}[a_{it}]$, which evolves according to

$$\mathbb{F}_{it}[a_{it}] = \hat{\rho}\mathbb{F}_{i,t-1}[a_{i,t-1}] + G(s_{it}^p - \hat{\rho}\mathbb{F}_{i,t-1}[a_{i,t-1}]). \tag{30}$$

Here, the perceived persistence $\hat{\rho}$ is allowed to differ from the actual persistence ρ, and the gain $G \in (0,1)$ can differ from the Bayes-rational Kalman gain G^*. The updating equation (30) nests a wide range of behavioral expectations, such as over-extrapolation (Angeletos et al. (2021), by setting $\hat{\rho} > \rho$), under-extrapolation or cognitive discounting (Gabaix (2020), by setting $\hat{\rho} < \rho$), and diagnostic expectations (by setting $\hat{\rho} = \rho$ and $G > G^*$). The following proposition characterizes the signs of own-decision and cross-decision predictability under common information and behavioral expectation (30):

Proposition 2. *Suppose both managers receive only the common, noisy productivity signal s_{it}^p in each period and use the behavioral belief in (30). Under the regularity conditions stated in Appendix A.7,*

- *if $\hat{\rho} > \rho$, then $\beta_\iota < 0$ and $\beta_h < 0$;*
- *if $\hat{\rho} < \rho$, then $\beta_\iota > 0$ and $\beta_h > 0$;*
- *if $\hat{\rho} = \rho$, then $sgn(\beta_h) = sgn(\beta_\iota) = sgn(G^* - G)$.*

Proof. See Appendix A.7. □

As we can see from Proposition 2, both misspecified persistence of productivity shock and non-Bayesian Kalman gains can lead to own- and cross-decision predictability. In the baseline scalar-fundamental model, however, common information makes the same behavioral distortion predict the investment forecast error through both past hiring and past investment. The investment forecast either overreacts or underreacts to both decisions, but cannot overreact to one and underreact to the other simultaneously. Proposition 3 below examines how this conclusion changes under factor-augmenting technology.

This same-sign pattern of own- and cross-decision predictability under common information may not hold under communication friction. If the labor manager has private information that the capital manager does not observe when making forecasts, on the right hand side of equation (29), the hiring forecast of the capital manager no longer coincides with the actual hiring of the labor manager in the period, and the covariance between productivity nowcast and hiring in $t-1$ is no longer zero. Hence, the rational investment forecast error can be predicted by lag hiring, i.e. $\beta_h^R \neq 0$. The following corollary characterizes the sign of β_h^R.

Corollary 1. *Suppose that information is segmented within the firm. If the policy coefficients in Lemma 1 satisfy $F_\iota^n > 0$, $F_\iota^a > 0$, and $F_h^a > 0$, then the cross-decision predictability of rational investment forecast error satisfies*

$$\beta_h^R \equiv \frac{\mathrm{Cov}(\iota_{it} - \mathbb{E}_{i,t-1}^k[\iota_{it}], h_{i,t-1})}{\mathrm{Var}(h_{i,t-1})} > 0.$$

Proof. See Appendix A.8. ◻

The conditions on the policy coefficients in Corollary 1 are easily satisfied under usual parameterizations. Under communication friction, the response of the capital manager's forecasts to the labor manager's decision is dampened because of the labor manager's private noisy information, making the investment forecast underreact to hiring in period $t-1$. As a result, the rational investment forecast error is positively correlated with past hiring. This is essentially the same belief-underreaction pattern that Coibion and Gorodnichenko (2012, 2015) highlight for noisy-information models.

Taken together, the communication friction within the firm makes it possible for the investment forecast to simultaneously overreact to investment (own decision) and underreact to hiring (other's decision). Having $\beta_\iota < 0$ is no longer sufficient to generate a negative cross-decision predictability β_h as in the common-information case. Under communication friction, there are two counteracting forces governing the sign of β_h:

$$\beta_h = \underbrace{\beta_h^R}_{>0,\text{ segmentation}} + \underbrace{\beta_h^B}_{<0,\text{ overreaction}} \tag{31}$$

On the one hand, cross-decision predictability inherits the same behavioral overreaction in expectations identified from own-decision predictability, which pushes β_h below zero as in Proposition 2. On the other hand, information segmentation from imperfect communication makes the investment forecast underreact to hiring, which pushes β_h above zero as in

Corollary 1. The sign of β_h depends on which force is stronger. If communication friction is strong enough, so that $\beta_h^R > -\beta_h^B$, then it is possible to observe $\beta_\iota < 0$ and $\beta_h > 0$: the investment forecast simultaneously overreacts to investment but underreacts to hiring.

Takeaway for the Empirics In sum, the joint sign pattern of regression coefficients β_ι and β_h in regressions (25) and (26) helps us identify communication friction in the data. The own-decision predictability β_ι is informative about the type of behavioral frictions in the capital manager's expectation, while the cross-decision predictability β_h contains additional information about whether or not the capital manager incorporates the labor manager's information in his information set. Specifically, Proposition 2 and Corollary 1 show that, within the baseline technology, the evidence for communication friction is strongest if we observe $\beta_\iota < 0 < \beta_h$ in the data, i.e. the investment forecast simultaneously overreacts to investment and underreacts to hiring.

3.2 Application to Data

I now apply the test (25) and (26) to a merged firm-level expectation-outcome dataset for US public firms.

Dataset For the expectation data, I use the IBES Guidance dataset, which contains all the guidance activities (i.e. the firm's public announcement on its own expectations about sales, capital expenditure, earnings per share, among many others, for the current fiscal year) of U.S. public firms since 1992. These guidances can be regarded as the self-reported expectation of the firm on its own financial performances. Since all measurements in the guidance are financial indicators (e.g. sales, capital expenditure, earnings per share, etc.), I assume that the expectations posted in IBES Guidance mainly reflect the capital manager's view about the firm's performance.

For the one-period-ahead investment forecast $\mathbb{F}^k_{i,t-1}[\iota_{it}]$, I use the first capital expenditure guidance posted by the firm in the first quarter of fiscal year t.[6] I then merge the IBES Guidance dataset to CRSP/Compustat Fundamental Annual and get a firm-level dataset with both expectations and realized outcomes. The details of data cleaning and sample construction are laid out in Appendix B.1. The resulting IBES-Compustat merged dataset

[6]Here, I regard capital expenditure (CPX) as a proxy for investment, although it is not exactly the same as ι_{it} in the model.

spans over 2003 and 2024 with 14,804 firm-year observations. Table 1 shows that the IBES-Compustat merged dataset is a sample for giant firms in the US economy: on average, firms in the merged sample is about two times larger than those in the CRSP-Compustat dataset in terms of sales and employment. The merged sample covers about 21.6% of total observations in the CRSP-Compustat Fundamental Annual Dataset between 2003 and 2024, and the total sales of firms in the merged IBES-Compustat sample take up about 29% of U.S. GDP in the sample period.

Table 1: Summary Statistics: Regression and Compustat Samples

Variable	Mean	Std. dev.	Median	Observations
Panel A: IBES/Compustat sample				
Sales	7,662.09	24,743.65	1,891.06	14,804
Gross property, plant, and equipment	6,235.37	23,815.35	1,157.65	14,768
Capital expenditures	499.86	1,831.10	88.89	14,804
Employment	21.84	69.21	6.40	14,804
Panel B: Full CRSP/Compustat Sample				
Sales	3,903.91	18,756.34	437.47	68,592
Gross property, plant, and equipment	2,762.06	15,765.27	172.64	65,413
Capital expenditures	230.50	1,382.78	13.20	68,592
Employment	11.43	53.99	1.30	68,592

Notes: Panel A is the IBES/Compustat sample used in the main expectation regressions. It requires a Q1 capital-expenditure forecast, nonmissing forecast error and lagged hiring, and positive lagged capital expenditures (14,804 firm-year observations, 2,418 firms, 2003–2024). Panel B is the CRSP/Compustat annual sample immediately before merging with IBES, restricted to the same year range as Panel A (68,592 firm-year observations, 7,567 firms, 2003–2024). It retains U.S.-incorporated firms reporting in U.S. dollars, excludes utilities (NAICS 22) and finance (NAICS 52), and requires positive sales and employment and nonmissing capital expenditures. Sales, gross property, plant, and equipment, and capital expenditures are in millions of nominal U.S. dollars; employment is in thousands.

For each firm-year observation in the IBES-Compustat merged sample, I take the realized capital expenditure from Compustat and impute the forecast/nowcast error as the arc-percent difference between the actual value and the predicted value, as in Davis and Haltiwanger (1992) and Bloom, Codreanu, and Fletcher (2025):

$$\text{CapEX Forecast Error}_{it} = \frac{\text{CapEX}_{it} - \mathbb{F}^k_{i,t-1}[\text{CapEX}_{it}]}{1/2(\text{CapEX}_{it} + \mathbb{F}^k_{i,t-1}[\text{CapEX}_{it}])}$$

The arc-difference definition of forecast errors has two advantages over the log difference. First, it is bounded between $[-2, 2]$, so that outliers in the realized values and expectations would not affect the forecast/nowcast error and the regression results by much. Additionally, it avoids the problems of taking logs when the actual values or expectations are close to 0. This may not be a big issue for the large-firm samples like IBES or Compustat because there are very few firm-year observations recording a zero capital expenditure or issuing a zero capital expenditure guidance. I verify that the regression results are robust to alternative definitions of forecast/nowcast errors (e.g. the log difference between actual and predicted values).

Regression Results Table 2 tabulates the regression result for the joint test of (25) and (26). The first column shows the result for own-decision regression (25). We can see that β_ι is negative and statistically significant, showing that the investment expectation of U.S. public firms systematically overreact to their own investment. This is consistent with the empirical findings in Barrero (2022) that sales expectations of firm managers in the U.S. present an overreaction pattern. From Proposition 2, this overreaction pattern can originate from over-extrapolation ($\hat{\rho} > \rho$) or over-responding to new signals ($G > G^*$, as in diagnostic expectation).

The second column shows the result for cross-decision regression (26). We can see that β_h is positive and statistically significant, which implies that the investment expectation of U.S. public firms systematically underreact to their hiring. From the discussions in Proposition 2 and Corollary 1, a negative own-decision predictability should predict a negative cross-decision predictability under perfect communication in the baseline scalar-fundamental model. The fact that own-decision and cross-decision predictability are in opposite directions indicates that the communication friction between capital and labor managers is strong enough to offset the channel of belief overreaction and overturn the sign of cross-decision predictability. The third column includes both lagged hiring and lagged investment in the regression, and these multivariate estimates are similar to the separate estimates in the first two columns. It shows that the correlation between past investment and past hiring does not bias the separate regression results in the first two columns.

Taken together, the results in Table 2 document that the investment forecasts of U.S. public firms simultaneously overreact to investment and underreact to hiring. This opposite-sign pattern rejects rational expectations through the nonzero β_ι and rejects common information within the baseline scalar-fundamental model through the positive β_h. The discussion below evaluates whether non-neutral technology can generate the same pattern under common information.

Table 2: Predictability of Investment Forecast Error by Past Hiring and Investment

	Dependent variable:		
	$\iota_t - \mathbb{F}^k_{t-1}[\iota_t]$		
	(1)	(2)	(3)
ι_{t-1}	-0.018^{**}		-0.022^{**}
	(0.009)		(0.009)
h_{t-1}		0.088^{***}	0.094^{***}
		(0.018)	(0.018)
Q1 Forecast Only	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes
Sector-year Fixed Effect	Yes	Yes	Yes
Observations	14,804	14,804	14,804
R^2	0.535	0.536	0.536

Notes: This table reports firm-level regressions of the capital-expenditure forecast error on lagged investment and lagged hiring. The forecast error is the arc-percent difference between realized capital expenditures and the firm's forecast. The sample uses the firm's first annual capital-expenditure guidance made in Q1 and requires nonmissing forecast errors and lagged hiring, as well as nonzero lagged capital expenditures. Lagged investment is the logarithm of capital expenditures in the preceding year, and lagged hiring is the preceding year's log change in employment. All specifications include firm and sector-by-year fixed effects, where sectors are defined using three-digit NAICS codes for manufacturing and two-digit NAICS codes otherwise. The sample contains 14,804 firm-year observations from 2,418 firms over 2003–2024. Standard errors clustered by firm are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Robustness Checks The first robustness exercise addresses the issue of omitted variable bias. The literature has documented that managerial forecast errors are persistent (Ma et al. (2020)). The decomposition in Proposition 1 also shows that lag nowcast error of productivity helps predict the investment forecast error. One may wonder whether the predictability of investment forecast error by past investment and past hiring in Table 2 are merely picking up the effects of these omitted lag forecast or nowcast error terms. To address this concern, I add lagged sales nowcast error as an additional control on the right-hand side of the regression (25) and (26). The sales nowcast error is defined as the arc difference between the actual sales in fiscal year t and the last sales guidance posted for fiscal year t. It serves as a proxy for the lag productivity nowcast error term in the decomposition of Proposition 1. The results are tabulated in Table 3. We can see that the estimates show similar pattern as Table 2: that the investment forecast overreacts to investment and underreacts to hiring. It confirms that the opposite-sign pattern we see in Table 2 is not an artifact of omitted lag forecast or nowcast errors.

Table 3: Robustness to Controls with Lag Forecast/Nowcast Errors

	Dependent variable:		
	$\iota_t - \mathbb{F}^k_{t-1}[\iota_t]$		
	(1)	(2)	(3)
ι_{t-1}	-0.033^{***}		-0.037^{***}
	(0.010)		(0.010)
h_{t-1}		0.096^{***}	0.104^{***}
		(0.022)	(0.023)
$y_{t-1} - \hat{y}^k_{t-1}$	0.113^{***}	0.089^{***}	0.095^{***}
	(0.034)	(0.033)	(0.033)
Firm Fixed Effect	Yes	Yes	Yes
Sector-Year Fixed Effect	Yes	Yes	Yes
Observations	8,455	8,455	8,455
R^2	0.562	0.562	0.563

Notes: This table reports regressions of the investment forecast error on lagged investment, lagged hiring, and the lagged sales nowcast error. The investment forecast error is defined as the arc difference between actual and expected capital expenditure, as in Table 2. The lagged sales nowcast error is defined analogously as the arc difference between actual sales in year $t-1$ and the last sales guidance issued before the end of year $t-1$. Firm and sector-year fixed effects are included as indicated. Standard errors clustered by firm are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

The other exercise concerns the robustness of the results to alternative definitions of the investment forecast error. I consider two alternatives: the log difference between actual and expected capital expenditure, and the level difference between actual and expected investment rate, defined as the ratio of capital expenditure in year t to the plants, properties and equipments (PPENT) in period $t-1$. I run the same regression (25) and (26) using these alternative forecast error measures and tabulate the results in Table 4. We can see that the opposite-sign pattern of own- and cross-decision predictability is robust to these alternative measures of forecast errors.

Table 4: Robustness to Alternative Definitions of Investment Forecast Errors

	(1) Arc difference	(2) Log difference	(3) Investment-rate difference
h_{t-1}	0.094^{***}	0.079^{***}	0.021^{***}
	(0.0185)	(0.0155)	(0.0040)
ι_{t-1}	-0.022^{**}	-0.027^{***}	-0.008^{***}
	(0.0086)	(0.0073)	(0.0018)
Firm fixed effects	Yes	Yes	Yes
Sector-year fixed effects	Yes	Yes	Yes
Q1 forecasts only	Yes	Yes	Yes
Observations	14,804	14,504	14,519
Firms	2,418	2,379	2,376
R^2	0.5362	0.4759	0.5146

Notes: This table reports joint regressions of alternative investment forecast-error measures on lagged hiring and lagged investment. In column (1), the dependent variable is the arc difference $(\text{CAPX}_{it} - F_{i,t-1}[\text{CAPX}_{it}]) / [0.5\,(\text{CAPX}_{it} + F_{i,t-1}[\text{CAPX}_{it}])]$. In column (2), it is $\log(\text{CAPX}_{it}) - \log(F_{i,t-1}[\text{CAPX}_{it}])$. In column (3), it is the investment-rate difference $(\text{CAPX}_{it} - F_{i,t-1}[\text{CAPX}_{it}]) / \text{PPENT}_{i,t-1}$, where PPENT denotes net property, plant, and equipment. Columns (2) and (3) trim observations below the 1st percentile or above the 99th percentile of their respective forecast-error measures. All specifications use Q1 forecasts and include firm and sector-year fixed effects. Standard errors, clustered by firm, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.3 Discussion

Alternative Common-information Explanations The opposite-sign pattern $\beta_\iota < 0 < \beta_h$ is inconsistent with the common-information benchmark characterized in Proposition 2, but it does not rule out other common-information model. To be precise, under rational expectation and common information, both past investment and past hiring should be contained in the information set of the capital manager, and neither should help predict the investment forecast error. This holds for any mechanisms of capital and labor adjustment. The relevant identification question is whether other mechanisms, when combined with behavioral overreaction, can cause lagged investment and lagged hiring to predict investment forecast errors in opposite directions even though managers share a common information set.

First, I consider the sign patterns of regression (25) and (26) under three alternative mechanisms: time-to-build in capital, Hsieh-Klenow type of distortions, and managers' private benefits. Each of these mechanisms can be accommodated by an extended version of our baseline model. As Appendix A.10.1, A.10.2 and A.10.3 shows, it turns out that, after taking log-quadratic approximation of the managers' payoff functions, the Markov Perfect

Equilibrium under these frictions take the same form as in Part 2 of Lemma 1. This means that the belief dynamics in Lemma 2 and the MPBE characterization in Lemma 3 remain true. These frictions affect investment and hiring by changing the policy matrix F_ι and F_h, and as long as the primitives of these frictions are common knowledge within the firm and do not violate the regularity conditions in Proposition 2, the same-sign pattern still holds for these alternative mechanisms under common information: without communication frictions, the cross-decision predictability β_h inherits the sign of the own-decision predictability β_ι. In other words, these frictions cannot reproduce opposite-sign pattern by themselves.

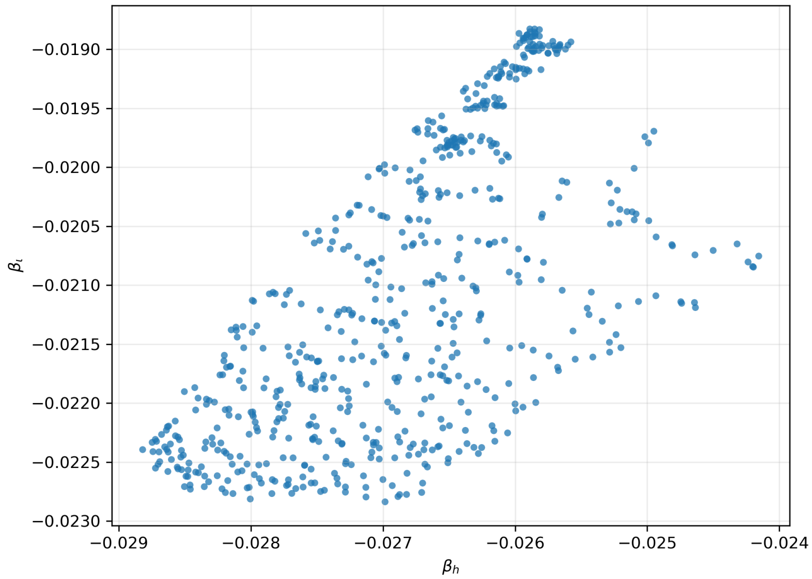
Next, I consider whether or not lumpiness in capital and labor adjustment can generate the opposite-sign pattern with belief overreaction. The same-sign result established in Proposition 2 is based on convex adjustment cost, which delivers small, continuous adjustment of capital and labor to the same direction in response to common productivity shocks. Lumpiness generated by non-convex adjustment costs (e.g. Cooper and Haltiwanger (2006), Asker et al. (2014)) can generate asynchronous adjustments between capital and labor and hence break the premise of Proposition 2.

I evaluate this possibility numerically by augmenting the baseline model with fixed adjustment costs on both capital and labor: if the capital manager makes a nonzero investment, he needs to pay a fixed fraction χ_k of total revenue as fixed costs; similarly, if the labor manager makes a nonzero hiring, she needs to pay a fixed fraction χ_n of total revenue as fixed costs. This setup makes fixed cost scales with firm size, as in Cooper and Haltiwanger (2006) and Bloom (2009). I shut off the information segmentation, let the managers learn from a common, noisy signal, and allow the managers' perceived persistence $\hat{\rho}$ to vary between ρ and 1. I solve the model numerically and simulate a panel of 1,000 firms for 500 periods to obtain the model-implied regression coefficients, with 625 combinations of non-convex costs $(\chi_k, \chi_n) \in [0, 0.01]$ and perceived persistence $\hat{\rho} \in (\rho, 1)$.

The results are scatter-plotted in Figure 1. We can see that for the parameter combination we consider, both own- and cross-decision predictability are negative. None of them generates the opposite-sign pattern $\beta_\iota < 0$ and $\beta_h > 0$ in the data. The result is not because the simulated model is short of lumpiness: the range of non-convex cost parameters we consider can generate inaction rate up to 50% for capital and labor; in comparison, the inaction rate of capital and labor adjustment in Compustat data is about 10%.[7] Although nonconvex costs can make capital and labor adjust at different dates, they are not sufficient to make capital and labor adjustment respond in opposite directions to the same productivity news.

[7] Here, I define inaction as investment or hiring rate below 10 percent, both in the data and in the model.

Figure 1: Simulated β_ι and β_h under Non-convex Costs & Over-extrapolation



Notes: This figure scatterplots the simulated own- and cross-decision predictability (β_ι, β_h) for 625 combinations of $(\chi_k, \chi_n, \hat{\rho}) \in [0, 0.01] \times [0, 0.01] \times (\rho, 1)$, equally spaced on each dimension, with $\sigma_{\epsilon,p} = 0.3$ and convex costs (ξ_k, ξ_n) fixed at their estimated value in Section 4.

As a result, if the capital manager's belief overreacts to investment, it should also overreact to hiring.

Lastly, I consider the implications of non-neutral technology for the sign pattern. To isolate this channel, I consider a flexible-input, common-information benchmark in which the firm's gross revenue is

$$Y_{it} = \left[\alpha^{1/\sigma}(A_{k,it}K_{it})^{1-1/\sigma} + (1-\alpha)^{1/\sigma}(A_{n,it}N_{it})^{1-1/\sigma}\right]^{\frac{\sigma(1-1/\epsilon)}{\sigma-1}}, \tag{32}$$

where $\epsilon > 1$ is the product-demand elasticity and $\sigma > 0$ is the capital–labor elasticity of substitution. The Cobb–Douglas case nested by $\sigma = 1$. $A_{k,it}$ and $A_{n,it}$ are capital-augmenting and labor-augmenting technologies, respectively. Their logged values follow two independent AR(1) process with a common persistence $\rho < 1$. The sign pattern of (β_ι, β_h) under non-neutral technology (32) is characterized in the following proposition.

Proposition 3 (Non-neutral technology under common information)**.** *Suppose the managers of the firm have common information and use Bayes-rational Kalman gain but with perceived persistence* $0 \leq \rho < \hat{\rho} < 1$. *Define the steady-state effective-capital share*

$$\bar{s}_K \equiv \frac{\alpha^{1/\sigma}(\bar{A}_k\bar{K})^{1-1/\sigma}}{\alpha^{1/\sigma}(\bar{A}_k\bar{K})^{1-1/\sigma} + (1-\alpha)^{1/\sigma}(\bar{A}_n\bar{N})^{1-1/\sigma}} \in (0,1). \tag{33}$$

Define the own-technology factor-demand responses

$$\psi_{kk} \equiv \bar{s}_K(\epsilon - 1) + (1 - \bar{s}_K)(\sigma - 1),$$
$$\psi_{nn} \equiv (1 - \bar{s}_K)(\epsilon - 1) + \bar{s}_K(\sigma - 1).$$

If $\psi_{kk} > 0$ and $\psi_{nn} > 0$, then, in the first-order log-linear approximation, the population coefficients in regressions (25) and (26) satisfy

$$\beta_\iota < 0, \qquad \operatorname{sgn}(\beta_h) = \operatorname{sgn}(\sigma - \epsilon). \tag{34}$$

Consequently,

$$\beta_\iota < 0 < \beta_h \quad \Longleftrightarrow \quad \sigma > \epsilon. \tag{35}$$

Proof. See Appendix A.9. □

The proposition states that, in order to get the opposite-sign pattern, we need strong substitutability between capital and labor inputs in the production function. When $\sigma > \epsilon$, the capital and labor are substitutes in the revenue function, in the sense that an increase in one input will lower the marginal revenue product of the other. In that case, a positive labor-augmenting technology shock increases hiring and decreases investment. When the capital manager's belief is over-extrapolative, he expects the labor-augmenting shock to last longer than the actual and lowers his investment forecast excessively. When the shock mean-reverts, actual investment exceeds the forecast, producing a larger forecast error and $\beta_h > 0$. In contrast, when $\sigma < \epsilon$, the capital and labor are complements in the revenue function: an increase in one input improves the marginal revenue product of the other input. In this case, a positive labor-augmenting technology shock will encourage both investment and hiring, and the own- and cross-decision predictability should also be in the same direction. The Cobb-Douglas production function I used in the baseline analysis is nested in this category.

How plausible is $\sigma > \epsilon$? The existing estimates from the empirical literature suggests that it's quite implausible. The estimates of complementarity/substitutability parameter σ are substantially below unity, suggesting that capital and labor are complements rather than substitutes in production: Raval (2019) reports estimates between 0.3 and 0.5, while Oberfield and Raval (2021) reports aggregate estimates between 0.5 and 0.7. In comparison, the demand elasticity parameter ϵ typically lies between 3 and 11 (e.g. Hottman, Redding, and Weinstein (2016)). Hence, non-neutral technology alone is empirically unlikely to account for the sign pattern that communication friction generates.

Taken together, these alternative mechanisms under common information do not readily reproduce the opposite-sign pattern. Frictions that preserve the common linear belief structure retain the same-sign restriction; non-convex costs do not overturn it numerically even with substantial lumpiness; and non-neutral technology requires $\sigma > \epsilon$, contrary to conventional elasticity estimates.

Communication Friction or Behavioral Friction? So far, I have been interpreting the cross-decision predictability $\beta_h > 0$ as communication friction that originates from management practices. An alternative interpretation is that β_h reflects another type of behavioral bias that the capital manager faces. Specifically, Lian (2020) and Giustinelli and Rossi (2023) illustrate that, when processing multi-dimensional information, households or managers think narrowly: they tend to truncate a complicated, multi-dimensional decision problem into smaller sub-problems and solve them individually, without fully internalizing the information that are more relevant in other sub-problems. In our setup, the fact that $\beta_h > 0$ while $\beta_\iota < 0$ can be because when capital manager is making forecast about his own future decisions, he views labor-side information as less relevant and hence pay less attention to it.

To distinguish a managerial friction from a behavioral friction requires a richer dataset that contains both managerial expectations and management practices.[8] If the cross-decision predictability is indeed driven by communication friction within the firm, then it should vary with different management practices across firms. For instance, whether the capital and labor decisions are made in the same place can be used as a proxy for communication friction. Given $\beta_\iota < 0$, if β_h is driven by communication friction, we should expect that β_h is more negative for the firms that report their capital and labor decisions are made in the same location while β_ι does not vary with the location. Instead, if β_h is driven by manager's internal behavioral bias, then it is not clear ex ante how the cross-decision predictability varies with the location. I leave this promising path of research for future work.

4 Quantitative Analysis

In this section, I quantify the aggregate effect of communication friction.

[8]An example is the Management and Organizational Practices Survey (MOPS) of U.S. Census Bureau. It elicits the plant-level expectations and management practices for 2015 and 2021 waves.

4.1 Calibration

An Extended Model for Quantification To obtain a credible quantification of the aggregate effect of communication friction, I extend our baseline model so that it incorporates Hsieh-Klenow-style reduced-form distortions and time-to-build in capital. The firm's profit function becomes

$$\Pi_{it} = (1-\tau_{it}^{y})R(K_{i,t-1}, N_{it}; A_{it}) - (1+\tau_{it}^{k})R_t K_{it} - W_t N_{it} - \frac{\xi_k}{2}\left(\frac{K_{it}}{K_{i,t-1}} - 1\right)^2 K_{i,t-1} - \frac{\xi_n}{2}\left(\frac{N_{it}}{N_{i,t-1}} - 1\right)^2 N_{i,t-1} \tag{36}$$

where $R(K_{i,t-1}, N_{it}; A_{it})$ is the revenue function with capital $K_{i,t-1}$ predetermined to capture time-to-build, and τ_{it}^{y} and τ_{it}^{k} are reduced-form firm-specific distortions as in Hsieh and Klenow (2009). τ_{it}^{y} captures the mechanisms that distort all factors symmetrically without changing the capital-labor ratio (such as price markup), while τ_{it}^{k} captures the capital-specific distortions (such as financial frictions) that generate variations in the capital-labor ratio. Following David and Venkateswaran (2019), I allow these reduced-form distortions to have a rich correlation structure:

$$\tau_{it}^{y} = \gamma^{y} a_{it} + \delta_i^{y} + \eta_{it}^{y} \tag{37}$$

$$\tau_{it}^{k} = \gamma^{k} a_{it} + \delta_i^{k} + \eta_{it}^{k} \tag{38}$$

where $\gamma^{y}, \gamma^{k} \in (0,1)$ captures correlation of distortions with productivity. They are referred to as "correlated distortion" as in Bartelsman et al. (2013) and capture size-dependent distortions (such as antitrust regulations). $\eta_{it}^{y} \sim N(0, \sigma_{\eta,y}^2)$ and $\eta_{it}^{k} \sim N(0, \sigma_{\eta,k}^2)$ are the transitory distortions uncorrelated to productivity, as in Hsieh and Klenow (2009). Lastly, $\delta_i^{y} \sim N(0, \sigma_{\delta,y}^2)$ and $\delta_i^{k} \sim N(0, \sigma_{\delta,k}^2)$ are firm characteristics that do not change over time. I assume that the correlated distortion γ^{y}, γ^{k}, firm characteristic $\delta_i^{y}, \delta_i^{k}$ and the uncorrelated distortions $\eta_{it}^{y}, \eta_{it}^{k}$ are common knowledge within the firm, so that the only source of uncertainty about τ^{y} and τ^{k} is the uncertainty about productivity a_{it}.

The information structure within the firm remains the same as in the baseline model: in each period, the capital manager and the labor manager receive a noisy, private signal about productivity a_{it} respectively, and they both receive a firm-level common signal. When they form expectations in each period, they do not know the other manager's decision in that period. To match the behavioral overreaction in own-decision predictability, I allow the expectations of capital manager to be over-extrapolative, i.e. the perceived persistence of productivity $\hat{\rho}$ is larger than the actual persistence ρ.

Calibration In total, there are 12 parameters to be calibrated in the extended model: convex adjustment costs ξ_k and ξ_n, the behavioral/information parameters $(\sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n}, \hat{\rho})$, and the parameters characterizing the reduced-form distortions $(\gamma^k, \gamma^y, \sigma_{\delta,k}, \sigma_{\delta,y}, \sigma_{\eta,k}, \sigma_{\eta,y})$. I pick 12 empirical moments as targets in the calibration. In particular, I include the own- and cross-decision predictability β_ι and β_h as targeted moments. As I have discussed in the previous section, the predictability of investment forecast errors by past investment and by past hiring are indicative of communication friction and belief extrapolation and hence help us identify the information and behavioral friction parameters. To match $\beta_h > 0$ and $\beta_\iota < 0$, we expect $\hat{\rho} > \rho$ and $\sigma_{\epsilon,k}, \sigma_{\epsilon,n} < \infty$, so that the private signals receive weights in belief updating. The rest of the targeted moments are standard in the firm dynamics literature. The serial correlation of input adjustments ($\text{corr}(\iota, \iota_{-1})$ and $\text{corr}(h, h_{-1})$) and the volatility of input adjustments ($var(\iota)$ and $var(h)$) are included to jointly discipline adjustment costs and uncorrelated distortions. The correlation between marginal revenue products and productivity, $\text{corr}(\text{MRPK}, a)$ and $\text{corr}(\text{MRPN}, a)$, is connected with the correlated distortions, and the dispersion in MRPK and MPRN allows the across-firm variation by the firm characteristic terms to absorb the proportion of observed MRPK and MRPN dispersion not captured by other variations.

The extended model is calibrated by minimum distance using analytical stationary moments. No finite panel is simulated in the calibration. For each candidate parameter vector, I proceed as follows:

1. For each $\Theta = (\xi_k, \xi_n, \hat{\rho}, \sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n}, \gamma^k, \gamma^y, \sigma_{\delta,k}, \sigma_{\delta,y}, \sigma_{\eta,k}, \sigma_{\eta,y})$, solve the MPBE according to Lemmas 1, 2, and 3, obtaining the linear policy rules for investment and hiring and the corresponding belief dynamics;

2. Stack the policy rules and belief dynamics into a linear state-space system. I compute the stationary covariance matrix of its dynamic component by solving the associated discrete Lyapunov equation and then add the cross-sectional covariance generated by the permanent distortions δ_i^k and δ_i^y analytically;

3. Construct the population model moments $\mathbf{m}(\Theta)$ directly from the resulting contemporaneous and lagged covariance matrices. In particular, variances, covariances, and correlations are computed from these matrices, and β_h and β_ι are the corresponding population projection coefficients.

Table 5: Estimated Parameter Values and Targets

Description		Value	Source/Target
Elasticity of substitution	ε	4	Standard Value
Capital share	α	0.47	Cost share in IBES/Compustat
Persistence of Productivity	ρ	0.94	Estimates of $a_{it} = \rho a_{i,t-1} + \mu_{it}$.
Std.dev. of Productivity shock	σ_μ	0.24	
Capital convex adj. cost	ξ_k	0.36	$\text{corr}(\iota, \iota_{-1})$
Labor convex adj. cost	ξ_n	3.75	$\text{corr}(h, h_{-1})$
Noise on the common signal	$\sigma_{\epsilon,p}$	1.40	Reg. Coef. β_ι
Noise on K manager's signal	$\sigma_{\epsilon,k}$	0.32	Reg. Coef. β_h
Noise on N manager's signal	$\sigma_{\epsilon,n}$	0.07	$\text{corr}(h, \Delta a)$
Degree of Extrapolation	$\hat{\rho}/\rho$	1.06	$\text{corr}(\iota, \Delta a)$
Correlated Distortion on revenue	γ_y	0.17	corr(MRPN, a)
Correlated Distortion on capital	γ_k	0.00	corr(MRPK, a)
Transitory Distortion η^y	$\sigma_{\eta,y}$	0.98	$\text{var}(\iota)$
Transitory Distortion η^k	$\sigma_{\eta,k}$	0.41	$\text{var}(h)$
Permanent Distortion δ^y	$\sigma_{\delta,y}$	1.06	var(MRPK)
Permanent Distortion δ^k	$\sigma_{\delta,k}$	1.36	var(MRPN)

I choose the parameter vector that minimizes the unweighted quadratic distance from the data moments $\mathbf{m}^{\text{data}}$:

$$\hat{\Theta} = \arg\min_{\Theta} \left[\mathbf{m}(\Theta) - \mathbf{m}^{\text{data}}\right]' \left[\mathbf{m}(\Theta) - \mathbf{m}^{\text{data}}\right].$$

Because the model moments are computed analytically, this criterion contains no model-simulation error. Appendix A.5 provides the details of the stationary-covariance calculation. The targeted moments and calibrated parameter values are summarized in Table 5.

Data The data moments are computed from a quantitative sample constructed by merging annual CRSP–Compustat records with IBES Guidance spanning from 2003 to 2024. From IBES, I obtain the first annual capital-expenditure guidance within the fiscal year with a forecast horizon below four quarters and compute the forecast error of investment by combining it with the actual capital expenditure in Compustat. From the Compustat sample, I take the book value of property, plant and equipment (PPE) as the measure of capital stock. Investment rate ι and hiring rate h are obtained by first-differencing the logged capital and employment series respectively. Value added is sales minus material expenditure.[9] I then

[9]Material expenditure is defined as Cost of Goods Sold (COGS) + Selling, General & Administrative Expenses (XSGA) − Depreciation (DP) − Wage Bill (XLR), following Keller and Yeaple (2009) and Flynn and Sastry (2020). For observations with missing Compustat labor expense, I impute the wage bill as

construct the total revenue factor productivity (TFPR) or sales Solow residual as the firm productivity a_{it} using time-to-build timing

$$a_{it} = \log(\text{Value Added}_{it}) - (1-\alpha)(1-1/\epsilon)n_{it} - \alpha(1-1/\epsilon)k_{i,t-1}$$

where the capital share α is chosen to match the cost share of labor in the sample, and the elasticity of substitution $\epsilon = 4$ is taken from the literature (Hsieh and Klenow (2009), Hottman et al. (2016)). The corresponding marginal-revenue-product proxies are $\text{MRPK}_{it} = \log(\text{Value Added}_{it}) - k_{i,t-1}$ and $\text{MRPN}_{it} = \log(\text{Value Added}_{it}) - n_{it}$.

Following David and Venkateswaran (2019), I remove industry-by-calendar-year fixed effects from investment, hiring, productivity growth and levels, MRPK, MRPN, and the capital–labor ratio $k_{i,t-1} - n_{it}$. Industries are defined by three-digit NAICS codes for manufacturing and two-digit NAICS codes otherwise; fiscal years ending from January through May are assigned to the preceding calendar year. Before trimming, I construct exact one-year changes in the residualized investment and hiring series. I then jointly retain observations between the first and ninety-ninth percentiles of the capital-expenditure forecast error, lagged capital expenditure, lagged hiring, those investment and hiring changes, and the residualized productivity, marginal-revenue-product, and capital–labor-ratio series. The resulting calibration sample contains 13,194 firm-year observations from 2,086 firms. Appendix B.2 provides the complete construction procedure and summary statistics of the sample.

4.2 Results

Communication Friction and Regression Coefficients Table 6 shows that our model can match most of the data moments closely, although it overstates the variance of investment and hiring. In particular, the model-implied regression coefficients β_h and β_ι closely match the data moments with over-extrapolation combined with a noisy firm-level common signal and more precise manager-level signals. In comparison, I mute communication frictions (i.e. fix $\sigma_{\epsilon,k}, \sigma_{\epsilon,n} \to \infty$), keep the common noisy signal and over-extrapolation and recalibrate the simplified model to the same set of targets. I am not able to match the sign and magnitude of the own- and cross-decision predictability in the data: the overextrapolative, common-information model gives us slightly positive, close-to-zero regression coefficients β_ι and β_h. This is consistent with the takeaway in Proposition 2 in Section 3.

employment multiplied by the corresponding industry-year wage from the Census Bureau County Business Patterns (CBP). See Appendix B.2 for details.

Table 6: Targeted Moments, Model vs Data

Moment	Data moment	Model-implied moment
$\text{corr}(\iota, \iota_{-1})$	0.305	0.306
$\text{corr}(h, h_{-1})$	0.164	0.168
$\text{corr}(\iota, \Delta a)$	0.160	0.159
$\text{corr}(h, \Delta a)$	0.302	0.303
β_h	0.112	0.108
β_ι	-0.026	-0.024
$\text{var}(\iota)$	0.019	0.036
$\text{corr}(\text{MRPK}, a)$	0.465	0.468
$\text{var}(\text{MRPK})$	0.455	0.456
$\text{var}(h)$	0.022	0.044
$\text{corr}(\text{MRPN}, a)$	0.465	0.468
$\text{var}(\text{MRPN})$	0.227	0.231

Sources of Learning The estimated parameters shed light on the learning patterns of the capital and labor managers. Table 7 shows how much the managers learn from multiple information sources. Repeatedly learning from noisy signals leads to a substantial reduction in uncertainty for both managers: posterior uncertainty is less than 10 percent of prior uncertainty for both the capital and labor manager. In this baseline calibration, managers' private signals play an important role in decision making. Both managers put a large weight on their private signals, so that the model can make investment forecast underreact to hiring and generate a positive cross-decision predictability. These private signals generate different posterior beliefs at the time investment and hiring decisions are made and break the perfect correlation between capital and labor managers' beliefs under perfect communication. In contrast, firm-level common signals are not major sources of learning for the managers: the Kalman gains on the common signals are below 2% for both managers. This finding is consistent with the literature: David et al. (2016) finds that firm-level public information, such as stock price, is not a major source of learning.

Communication Friction as Input-specific Distortions Communication friction plays an important role in two untargeted moments: the investment-hiring correlation and the dispersion of capital-labor ratio. These two moments show that communication frictions affect firm dynamics in a significantly different manner from other behavioral and informational frictions.

Low investment-hiring correlation is an important feature in the US firm data, and our model

Table 7: Learning Patterns of the Managers

	Uncertainty		Source of Learning		
Manager	Posterior	Posterior/Prior	Common	Private	Prior
Capital Manager	0.035	7.85%	1.81%	34.33%	63.86%
Labor Manager	0.005	1.11%	0.26%	90.71%	9.03%

Notes: "Posterior" reports the posterior variance $\hat{\Sigma}^m$ for manager $m \in \{k, n\}$. "Posterior/Prior" reports $\hat{\Sigma}^m/\sigma_a^2$, the posterior variance relative to the unconditional variance of productivity. The Common, Private, and Prior columns report the weights in the manager's posterior mean on the firm-level common signal s_{it}^p, the manager-specific private signal s_{it}^m, and the common prior $\bar{a}_{it}$, respectively.

provides a decent match as an untargeted moment. Which mechanisms are responsible for it, and which one is the most prominent? In the first column of Table 8, I report the contribution of each friction to the investment-hiring correlation. The numbers in the table are computed under the assumption that only one friction of interest is active (i.e. the relevant parameters are fixed at the estimated level) while all other frictions are muted (i.e. the relevant parameters are set to the frictionless level). From Table 8, we can see that the low investment-hiring correlation is primarily driven by communication friction and time-to-build, with communication friction being the strongest force in the model: communication friction alone generates a 0.33 investment-hiring correlation, while none of the other frictions alone lowers the correlation effectively from 1.

The decomposition of investment-hiring correlation highlights a crucial difference between firm-level, common information/behavioral frictions and within-firm segmented information. We can see that if we remove the information segmentation and allow the managers to learn from all three signals received in each period, the investment-hiring correlation is equal to 1. Manager-level private signals, on the other hand, lower the correlation significantly. This can be seen from the expressions of capital and labor choices. Without other frictions, capital and labor are both proportional to the managers' posterior beliefs about productivity, i.e. $k_{it} \propto \mathbb{E}_{it}^k[a_{it}]$ and $n_{it} \propto \mathbb{E}_{it}^n[a_{it}]$. Absent communication friction, managers share the same information, so that $\mathbb{E}_{it}^k = \mathbb{E}_{it}^n = \mathbb{E}_{it}$, and firm-level common noisy information attenuates the responsiveness of both inputs by the same proportion, without breaking the perfect correlation between them. The existence of private signals breaks this pattern as the managers' expectations are no longer perfectly correlated in general.

The distinction between firm-level common information and manager-level private information is akin to the difference between revenue distortion τ^y and capital distortion τ^k. Common, noisy information and behavioral expectations act like the revenue distortion τ^y as they both affect all the inputs symmetrically, while the information segmentation introduced by communication friction acts like the capital-specific distortion τ^k, as they both affect different inputs disproportionately and generate variations in capital-labor ratio. Communication frictions should therefore be categorized as input-specific distortions, while common noisy information and behavioral expectations belong to the class of revenue distortion.

To make the point clearer, I investigate the role that communication friction plays in the dispersion of capital-labor ratio. As we know from Hsieh and Klenow (2009), revenue distortions cannot distort capital-labor ratio while input-specific distortions can. The second column of Table 8 reports the capital-labor ratio dispersion that each mechanism can generate alone. We can see that in our calibrated model, communication friction is an important source of the capital-labor ratio variability. Communication friction alone generates a capital-labor ratio dispersion of 0.302, which accounts for approximately 39% of the observed dispersion in the capital-labor ratio and about 38% of the model-implied dispersion. In contrast, revenue distortion τ^y, common noisy information and behavioral expectations cannot distort the capital-labor ratio by themselves. These results show that modeling information and behavioral friction as commonly shared within the firm can be misleading. It ignores an important channel for imperfect information to generate input-specific distortions, which may lead us to underestimate the role of information and behavioral friction in firm dynamics and overestimate the role of other input-specific distortions (e.g. financial frictions, non-neutral technologies) in driving the variations of capital-labor ratio.

Table 8: Sources of input comovement and capital–labor ratio variation

Mechanism	$\text{corr}(\iota_t, h_t)$	$\text{var}(k_{t-1} - n_t)$
Data	0.557	0.774
Model, all mechanisms	0.445	0.802
Time to build only	0.631	0.085
Convex costs only	0.993	0.006
Communication friction only	0.326	0.302
Overextrapolation + common information	1.000	0.000
Capital distortion τ^k only	0.972	2.032
Revenue distortion τ^y only	1.000	0.000

Notes: "Only" rows activate the named mechanism at its estimated value and mute all other mechanisms. The τ^k-only row keeps γ_k, $\sigma_{\delta,k}$, and $\sigma_{\eta,k}$ active; the τ^y-only row keeps γ_y, $\sigma_{\delta,y}$, and $\sigma_{\eta,y}$ active. The common-information counterfactual combines the three estimated signal precisions into one common signal.

Table 9: Misallocation accounting

Mechanism	var(MRPK)	var(MRPN)	$\log A - \log A^*$
Data	0.455 (100.00%)	0.227 (100.00%)	–
Model, all mechanisms	0.456 (100.36%)	0.231 (101.70%)	-37.54%
Time to build only	0.085 (18.79%)	0.000 (0.00%)	-4.91%
Convex costs only	0.029 (6.45%)	0.058 (25.69%)	-8.68%
Communication friction only	0.123 (27.10%)	0.044 (19.28%)	-4.99%
Overextrapolation + common information	0.003 (0.59%)	0.003 (1.18%)	-0.54%
Correlated distortions only	0.013 (2.89%)	0.013 (5.75%)	-2.62%
Transitory distortions η	0.012 (2.56%)	0.018 (7.86%)	-1.01%
Permanent distortions δ	0.316 (69.37%)	0.152 (67.16%)	-18.69%

Notes: Dispersion shares in parentheses use the corresponding data MRPK and MRPN dispersion as the denominator. "Only" rows activate the named mechanism at its estimated value and mute the others. Permanent and transitory distortion rows report the full-model value minus the counterfactual that removes that distortion block. Aggregate TFP loss entries are log TFP gaps relative to the frictionless allocation.

Allocative Efficiency What is the implication of communication frictions for input misallocation and allocative efficiency? In Table 9, I tabulate the contribution of each friction on the capital and labor misallocation and quantify the aggregate TFP loss generated by each friction. These numbers are computed under the assumption that only one friction is active each time. We can see that communication friction alone can generate approximately 27 percent of observed MRPK dispersion and 20% of MRPN dispersion. In comparison, if I mute communication friction within the firm, the MRPK and MRPN dispersion generated by behavioral expectations and common noisy information are much smaller. Hence, the information-driven misallocation is primarily generated by communication friction. Compared with other mechanisms, the MRPK and MRPN dispersion driven by communication frictions are quantitatively significant: neither adjustment costs nor correlated distortions can generate a significantly larger marginal revenue product dispersion than communication frictions by themselves.

Despite the fact that communication friction can generate a large portion of MRPK and MRPN dispersion, it is not the largest source of aggregate TFP loss. Communication friction by itself generates a sizable 5% of aggregate TFP loss from the frictionless benchmark, but its magnitude is smaller than adjustment costs, which explains much less MRPK dispersion than communication frictions. This is because the MRPK and MRPN dispersion are not sufficient statistics for aggregate TFP loss by themselves. Analytically, the aggregate TFP loss can be expressed as

$$\log(A_t) - \log(A^*) = -\frac{1}{2}\frac{\epsilon^2}{\epsilon - 1}\Big[\hat{\alpha}_1(1-\hat{\alpha}_2)\text{var}(\text{MRPK}) + \hat{\alpha}_2(1-\hat{\alpha}_1)\text{var}(\text{MRPN}) + 2\hat{\alpha}_1\hat{\alpha}_2 \cdot \text{cov}(\text{MRPK, MRPN})\Big] \tag{39}$$

where $\hat{\alpha}_1$ and $\hat{\alpha}_2$ are the capital and labor elasticities in the revenue function. The covariance term captures how the correlation between capital and labor distortions affect aggregate productivity. Intuitively, if capital distortion and labor distortion comove with each other, the distorted capital choice would distort the labor choice even further due to the complementarity between capital and labor, which leads to larger aggregate TFP loss. If the capital distortion and labor distortion are negatively correlated, this reinforcement effect disappears, making the resulting aggregate TFP loss smaller.

Table 10: Comovement between marginal revenue products

Mechanism	corr(MRPK, MRPN)
Data	-0.144
Model, all mechanisms	-0.177
Time to build only	-
Convex costs only	0.987
Communication friction only	-0.919
Overextrapolation + common information	1.000
Capital distortion τ^k only	-
Revenue distortion τ^y only	1.000

Notes: Mechanism-only rows activate the named mechanism at its estimated value and mute all other mechanisms. The correlation is undefined for time-to-build-only and τ^k-only because these counterfactuals generate no MRPN dispersion. The τ^k-only row keeps γ_k, $\sigma_{\delta,k}$, and $\sigma_{\eta,k}$ active; the τ^y-only row keeps γ_y, $\sigma_{\delta,y}$, and $\sigma_{\eta,y}$ active. The common-information counterfactual combines the three estimated signal precisions into one common signal.

We can see from Table 10 that the mechanisms in the calibrated model present different patterns of marginal revenue product correlation. Specifically, communication friction generates a negative correlation between marginal revenue product, time-to-build and capital distortion generate a zero correlation, and other mechanisms generate a positive correlation. This explains why communication frictions imply a smaller aggregate TFP loss than convex adjustment costs even though the latter explains a smaller proportion of marginal revenue product dispersion.

But why are MRPK and MRPN negatively correlated under communication friction? It's because under communication friction, the comovement of MRPK and MRPN is primarily driven by manager-level private signals. Suppose there is a favorable realization of the capital manager's private signal while the productivity of the firm remains the same. Upon observing the private signal, the capital manager updates his beliefs and invests in capital, but since the productivity does not change, the marginal revenue product of capital decreases due to diminishing marginal return. Meanwhile, a higher capital improves the marginal revenue product of labor due to complementarity. Therefore, a favorable realization of the capital manager's private signal leads to a low MRPK and a high MRPN. The same mechanism holds for the labor manager's private signal as well. Since both capital and labor managers rely heavily upon their own private signals as in Table 7, the MRPK and MRPN are negatively correlated.

In contrast, other mechanisms considered in the model cannot generate a negative correlation between MRPK and MRPN by themselves. Under time-to-build or capital distortion τ^k, labor decisions are frictionless, and hence the MRPN dispersion is zero. Under common information, behavioral expectation or revenue distortion τ^y, both MRPK and MRPN are affected symmetrically, and the correlation is perfect without other frictions. Under adjustment costs, both investment and hiring respond sluggishly to the same productivity shock, and hence the marginal revenue products tend to comove in the same direction. Hence, communication friction appears to be an important force to make the model-implied MRP correlation negative and close to the untargeted data moment.

In sum, although communication friction generates a large share of observed MRPK and MRPN dispersion alone, the aggregate TFP loss from this mechanism is mitigated by the negative correlation between marginal revenue product of capital and labor. More generally, this shows the importance of investigating the nature of distortions in misallocation and aggregate productivity accounting: for the frictions that move MRPK and MRPN in opposite directions, such as heterogeneous technology in David and Venkateswaran (2019) and communication friction, large dispersions in marginal revenue products do not automatically imply large losses in TFP.

Discussion My quantification of information-driven misallocation is comparable to the efficiency loss suggested by the literature. David et al. (2016) estimate the US aggregate TFP loss from information friction to be between 4% (in the case of frictionless labor) and 40% (when labor is subject to the same noisy information as capital). David and Venkateswaran (2019)'s estimate is around 7%. My quantification is within the range and closer to the lower bound. Compared with the papers that treat information friction as a revenue distortion, such as Gautier et al. (2025) and Ropele et al. (2024), my quantification is larger.

The takeaways for other frictions are largely consistent with the literature. As in David and Venkateswaran (2019), permanent distortions δ^k and δ^y remain the largest contributors to the capital and labor misallocation and aggregate TFP loss. My current framework focuses on within-firm friction and assumes that the internal information structure is same across firms. Since management and organizational practices are highly sticky over time and heterogeneous across firms (Bloom et al. (2019)), they can be candidates of these firm-level permanent distortions. A promising direction for future research is to allow for heterogeneous practices of internal information sharing across firms and evaluate its quantitative plausibility as a candidate of permanent distortions. For instance, in the Census MOPS dataset, I can use decision locations of capital and labor discussed to proxy internal information sharing: I can treat the same-location group as having common information for capital and labor, and the different-location group as having segmented information. I can then examine whether it accounts for the persistent firm-level components of MRPK and MRPN. This would provide a way to open the black box of the permanent distortion terms and quantify how much of their dispersion comes from this channel of internal information sharing.

Lastly, the misallocation accounting exercise should not be treated as a welfare analysis. In the current framework, the firm's internal information structure is exogenously given, and information segmentation is strictly dominated by perfect communication: making the managers' private signals common knowledge within the firm will always improve the value of the firm and generate a net TFP gain in aggregate. In reality, the internal information flow might be endogenously chosen by the firm. For instance, if each manager has limited attention and the complementarity of tasks between managers is weak, then it might be optimal for the firm to take the attention constraint into account and assign segmented information to managers (Dessein et al. (2016)). This might change the welfare implications of this paper: having segmented information within the firm may not be interpreted as a "misallocation" of information but a second-best outcome. I leave this to a companion paper that studies information sharing across plants within multi-unit firms.

5 Conclusion

This paper studies how imperfect information sharing inside firms affects managerial expectations, input choices, and allocative efficiency. I develop a dynamic team-production framework in which capital and labor managers pursue the same objective but make decisions using different information. The framework produces a novel empirical test based on investment forecast errors. Under common information, behavioral overreaction or underreaction makes forecast errors predictable by past investment and past hiring in the same direction. Information segmentation breaks this restriction because hiring contains information observed by the labor manager but not fully incorporated into the capital manager's forecast. Applying the test to the data on U.S. public firms, I find that investment forecasts simultaneously overreact to investment and underreact to hiring. This pattern is difficult to reconcile with the common-information mechanisms considered in the paper, including extrapolative beliefs, convex and non-convex adjustment costs, reduced-form distortions, and empirically plausible non-neutral technologies. The evidence is therefore inconsistent with the common-information benchmark and supports segmented information within firms.

The quantitative results show why this distinction matters for firm dynamics. Manager-specific information substantially lowers investment–hiring comovement and acts as an input-specific distortion, unlike common noisy information and behavioral expectations, which affect capital and labor symmetrically. Communication frictions alone can generate approximately 40% of observed capital–labor-ratio dispersion, 27% of MRPK dispersion, and 20% of MRPN dispersion. They imply an aggregate TFP loss of about 5% relative to the frictionless allocation. The negative MRPK–MRPN correlation generated by manager-specific private information attenuates the associated TFP loss relative to what the MRPK and MRPN dispersions alone would imply.

The broader implication is that firm-level information frictions cannot always be represented as a single imperfect expectation shared throughout the organization. How information is distributed across decision-makers has important implications for firm dynamics and allocative efficiency. Linking the forecast-error test developed here to direct measures of delegation, communication, and decision location—and allowing information structures to vary across firms—offers a natural next step for identifying the organizational origins of information segmentation and its aggregate consequences.

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A Theory Appendix

A.1 Second-order Approximation of the Profit Function

Denote the frictionless steady state value of variable X_t as $\bar{X}$, and denote the log deviation of X_t from $\bar{X}$ as x_t.

Frictionless Steady State Fix $\bar{A} = 1$. The profit maximization problem without information and adjustment frictions is to choose K and N to maximize

$$\Pi(K, N) = Y^{1/\epsilon} A^{1-1/\epsilon} (K^{\alpha} N^{1-\alpha})^{1-1/\epsilon} - RK - WN$$

FOC:

$$Y^{1/\epsilon} A^{1-1/\epsilon} K^{\alpha(1-1/\epsilon)-1} N^{(1-\alpha)(1-1/\epsilon)} \alpha(1 - 1/\epsilon) = R$$

$$Y^{1/\epsilon} A^{1-1/\epsilon} K^{\alpha(1-1/\epsilon)} N^{(1-\alpha)(1-1/\epsilon)-1} (1 - \alpha)(1 - 1/\epsilon) = W$$

In the quantitative exercise, $\alpha, \epsilon, \rho, \sigma_u, \delta, \beta$ are calibrated outside the model. Given these parameters, the frictionless steady-state values for rental rate, capital, labor and wage can be expressed analytically. First, rental price $\bar{R}$ is

$$\bar{R} = 1/\gamma - 1 + \delta = \alpha(1 - 1/\epsilon) \bar{K}^{\alpha(1-1/\epsilon)-1} N^{(1-\alpha)(1-1/\epsilon)}$$

Normalize $\bar{Y} = 1, \bar{A} = 1$, we get

$$\bar{K}^{\alpha(1-1/\epsilon)} \bar{N}^{(1-\alpha)(1-1/\epsilon)} = 1$$

We can use the above two equations, together with FOCs, to back out steady state $\bar{K}, \bar{N}, \bar{W}$:

$$\bar{K} = \frac{\alpha(1 - 1/\epsilon)}{1/\gamma - 1 + \delta}$$

$$\bar{N} = \left(\frac{1/\gamma - 1 + \delta}{\alpha(1 - 1/\epsilon)} \right)^{\alpha/(1-\alpha)}$$

$$\bar{W} = \bar{K}^{\alpha} \bar{N}^{-\alpha} (1 - \alpha)(1 - 1/\epsilon)$$

Second-order Approximation We have

$$
\begin{aligned}
\Pi \approx\ & \bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)} + \bar{R}\bar{K}k_t + \bar{W}\bar{N}n_t \\
& + \frac{1}{2}\bar{R}\bar{K}(1-1+\alpha(1-1/\epsilon))k_t^2 + \frac{1}{2}\bar{W}\bar{N}(1-1+(1-\alpha)(1-1/\epsilon))n_t^2 \\
& + \bar{A}^{1-1/\epsilon}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}k_t n_t \\
& - \bar{R}\bar{K} - \bar{R}\bar{K}k_t - \bar{R}\bar{K}\frac{1}{2}k_t^2 \\
& - \bar{W}\bar{N} - \bar{W}\bar{N}n_t - \bar{W}\bar{N}\frac{1}{2}n_t^2 \\
& - \frac{\xi_k}{2}\bar{K}(k_t - k_{t-1})^2 - \frac{\xi_n}{2}\bar{N}(n_t - n_{t-1})^2 \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}(a_t + \frac{1}{2}a_t^2) - \frac{1}{2}\frac{\epsilon-1}{\epsilon^2}\bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}a_t^2 \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}\alpha(1-1/\epsilon)\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}k_t a_t \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}(1-\alpha)(1-1/\epsilon)\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}n_t a_t
\end{aligned}
$$

where the sixth line is an approximation of adjustment cost for capital:

$$
\begin{aligned}
\frac{\xi_k}{2}\left(\frac{K_t - (1-\delta)K_{t-1}}{K_{t-1}} - \delta\right)^2 K_{t-1} &= \frac{\xi_k}{2}\left(\exp(k_t - k_{t-1}) - 1\right)^2 \bar{K}\exp(k_{t-1}) \\
&= \frac{\xi_k \bar{K}}{2}(k_t - k_{t-1})^2(1 + k_{t-1} + ...) = \frac{\xi_k \bar{K}}{2}(k_t - k_{t-1})^2
\end{aligned}
$$

Define investment $\iota_t \equiv k_t - k_{t-1}$ and hiring $h_t \equiv n_t - n_{t-1}$. Replace $k_t = k_{t-1} + \iota_t$ and $n_t = h_t + n_{t-1}$, we get:

Hence

$$
\begin{aligned}
\pi =\ & \frac{1}{\epsilon}\bar{Y} + \frac{\epsilon-1}{\epsilon}\bar{Y}a_t + \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y}a_t^2 \\
& + \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1)\left(k_{t-1}^2 + 2k_{t-1}\iota_t + \iota_t^2\right) \\
& + \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1)(n_{t-1}^2 + 2n_{t-1}h_t + h_t^2) \\
& + \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}(k_{t-1}n_{t-1} + \iota_t n_{t-1} + k_{t-1}h_t + \iota_t h_t) \\
& + \frac{\epsilon-1}{\epsilon}\alpha(1-1/\epsilon)\bar{Y}(k_{t-1}a_t + \iota_t a_t) + \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y}(n_{t-1}a_t + h_t a_t) \\
& - \frac{\xi_k}{2}\bar{K}\iota_t^2 - \frac{\xi_n}{2}\bar{N}h_t^2
\end{aligned}
\tag{A1}
$$

Hence the log-quadratic profit function takes the form

$$x_t' P x_t + x_t' Q \iota_t + x_t' R h_t + H_\iota \iota_t^2 + H_{\iota h} \iota_t h_t + H_h h_t^2 \tag{A2}$$

where state vector $x_t = [1, k_{t-1}, n_{t-1}, a_t]$, and the matrices are

$$P = \begin{bmatrix} \frac{1}{\epsilon}\bar{Y} & 0 & 0 & \frac{\epsilon-1}{2\epsilon}\bar{Y} \\ * & \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) & \frac{1}{2}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} & \frac{\epsilon-1}{2\epsilon}\alpha(1-1/\epsilon)\bar{Y} \\ * & * & \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) & \frac{\epsilon-1}{2\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y} \\ * & * & * & \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y} \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 \\ \bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \frac{\epsilon-1}{\epsilon}\alpha(1-1/\epsilon)\bar{Y} \end{bmatrix}$$

$$R = \begin{bmatrix} 0 \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) \\ \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y} \end{bmatrix}$$

and $H_\iota = \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) - \frac{1}{2}\xi_k\bar{K}$, $H_h = \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) - \frac{1}{2}\xi_n\bar{N}$, $H_{\iota h} = \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}$.

The law of motion for the state vector x_t is given by

$$x_{t+1} \equiv \begin{bmatrix} 1 \\ k_{t-1} + \iota_t \\ n_{t-1} + h_t \\ \rho a_t + \mu_{t+1} \end{bmatrix} = A x_t + B \iota_t + C h_t + D \mu_{t+1}$$

where

$$A \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \rho \end{bmatrix}, B \equiv \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, C \equiv \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, D \equiv \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

A.2 Proof of Lemma 2

Let

$$\Sigma^- \equiv \rho^2 \bar{\Sigma} + \sigma_\mu^2.$$

At the beginning of period t, before the current signals arrive, the information revealed at the end of period $t-1$ is common to both managers. Hence their common prior is

$$a_{it} \mid \mathcal{P}_{it} \sim N\big(\bar{a}_{it}, \Sigma^-\big),$$

where $\mathcal{P}_{it}$ denotes this common information.

We first derive the managers' first-order beliefs. Conditional on $\mathcal{P}_{it}$, the prior and all signal noises are Gaussian and independent. The capital manager observes s_{it}^p and s_{it}^k. Normal–normal updating therefore gives

$$\begin{aligned} \left(\hat{\Sigma}^k\right)^{-1} &= \left(\Sigma^-\right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2}, \\ \hat{a}_{it}^k &= \hat{\Sigma}^k \left[\left(\Sigma^-\right)^{-1} \bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^p + \sigma_{\epsilon,k}^{-2} s_{it}^k\right]. \end{aligned}$$

The same calculation for the labor manager yields

$$\begin{aligned} \left(\hat{\Sigma}^n\right)^{-1} &= \left(\Sigma^-\right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,n}^{-2}, \\ \hat{a}_{it}^n &= \hat{\Sigma}^n \left[\left(\Sigma^-\right)^{-1} \bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^p + \sigma_{\epsilon,n}^{-2} s_{it}^n\right]. \end{aligned}$$

Substituting the definition of Σ^- gives equations (19) and (20).

Next consider the capital manager's expectation of the labor manager's posterior. The only term in $\hat{a}_{it}^n$ that is not measurable with respect to $\mathcal{I}_{it}^k$ is s_{it}^n. Since $s_{it}^n = a_{it} + \epsilon_{it}^n$, with ϵ_{it}^n independent of $\mathcal{I}_{it}^k$,

$$\mathbb{E}\big[s_{it}^n \mid \mathcal{I}_{it}^k\big] = \mathbb{E}\big[a_{it} \mid \mathcal{I}_{it}^k\big] = \hat{a}_{it}^k.$$

Taking the conditional expectation of the labor manager's posterior formula therefore gives

$$\hat{a}_{it}^{(k,n)} = \hat{\Sigma}^n \left[\left(\Sigma^-\right)^{-1} \bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^p + \sigma_{\epsilon,n}^{-2} \hat{a}_{it}^k \right],$$

which is equation (21). Reversing the roles of the two managers gives

$$\hat{a}_{it}^{(n,k)} = \hat{\Sigma}^k \left[\left(\Sigma^-\right)^{-1} \bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^p + \sigma_{\epsilon,k}^{-2} \hat{a}_{it}^n \right],$$

which is equation (22). For completeness, the corresponding conditional variances are

$$\begin{aligned} \hat{\Sigma}^{(k,n)} &= \left(\hat{\Sigma}^n \sigma_{\epsilon,n}^{-2}\right)^2 \left(\hat{\Sigma}^k + \sigma_{\epsilon,n}^2\right), \\ \hat{\Sigma}^{(n,k)} &= \left(\hat{\Sigma}^k \sigma_{\epsilon,k}^{-2}\right)^2 \left(\hat{\Sigma}^n + \sigma_{\epsilon,k}^2\right). \end{aligned}$$

Thus the second-order beliefs are obtained directly from the first-order posteriors; no recursive covariance state is required.

It remains to establish the common-prior recursion. At the end of period t, the actions and predetermined inputs are public. Lemma 1 and $F_\iota^a F_h^a \neq 0$ imply

$$\begin{aligned} \hat{a}_{it}^k &= \frac{\iota_{it} - F_\iota^k k_{i,t-1} - F_\iota^n n_{i,t-1}}{F_\iota^a}, \\ \hat{a}_{it}^n &= \frac{h_{it} - F_h^k k_{i,t-1} - F_h^n n_{i,t-1}}{F_h^a}. \end{aligned}$$

Hence both posterior means become common knowledge. The posterior formulas above can then be inverted to recover both private signals:

$$\begin{aligned} s_{it}^k &= \sigma_{\epsilon,k}^2 \left[\left(\hat{\Sigma}^k\right)^{-1} \hat{a}_{it}^k - \left(\Sigma^-\right)^{-1} \bar{a}_{it} - \sigma_{\epsilon,p}^{-2} s_{it}^p \right], \\ s_{it}^n &= \sigma_{\epsilon,n}^2 \left[\left(\hat{\Sigma}^n\right)^{-1} \hat{a}_{it}^n - \left(\Sigma^-\right)^{-1} \bar{a}_{it} - \sigma_{\epsilon,p}^{-2} s_{it}^p \right]. \end{aligned}$$

After pooling the public and private signals, the common end-of-period posterior mean is

$$\tilde{a}_{it} = \bar{\Sigma} \left[\left(\Sigma^-\right)^{-1} \bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^p + \sigma_{\epsilon,k}^{-2} s_{it}^k + \sigma_{\epsilon,n}^{-2} s_{it}^n \right],$$

and its variance is

$$\bar{\Sigma} = \left[\left(\Sigma^-\right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2} + \sigma_{\epsilon,n}^{-2} \right]^{-1}.$$

Because $a_{i,t+1} = \rho a_{it} + \mu_{i,t+1}$, the next common prior has mean $\bar{a}_{i,t+1} = \rho \tilde{a}_{it}$ and variance

$\rho^2\bar{\Sigma} + \sigma_\mu^2 = \Sigma^-$. Re-indexing this mean recursion gives equation (23), and stationarity gives the fixed-point equation for $\bar{\Sigma}$ stated in the lemma.

A.3 Proof of Lemma 1

A.3.1 Part I: linearity of MPE

Guess that ι and h are linear:

$$\iota_t = F_\iota x_t, h_t = F_h x_t$$

Hence, the MPE is defined as

- Given F_h, the capital manager's strategy $\iota_t = F_\iota x_t$ solves the Bellman equation

$$\begin{aligned} V^k(x_t) &= \max_{\iota_t} x_t' P x_t + x_t' Q \iota_t + x_t' R F_h x_t + H_\iota \iota_t^2 + H_{\iota h} \iota_t F_h x_t + x_t' F_h' H_h F_h x_t + \gamma E\left[V^k(x_{t+1})\right] \\ &= \max_{\iota_t} x_t' \left(P + R F_h + F_h' H_h F_h\right) x_t + x_t' \left(Q + F_h' H_{\iota h}\right) \iota_t + H_\iota \iota_t^2 + \gamma E\left[V^k(x_{t+1})\right] \end{aligned} \tag{A3}$$

  subject to the law of motion (15) with h_t replaced by $F_h x_t$:

$$x_{t+1} = \left(A + C F_h\right) x_t + B \iota_t + D \mu_{i,t+1} \tag{A4}$$

- Given F_ι, the labor manager's strategy $h_t = F_h x_t$ solves the Bellman equation

$$\begin{aligned} V^n(x_t) &= \max_{h_t} x_t' P x_t + x_t' Q F_\iota x_t + x_t' R h_t + x_t' F_\iota' H_\iota F_\iota x_t + x_t' F_\iota' H_{\iota h} h_t + H_h h_t^2 + \gamma E\left[V^n(x_{t+1})\right] \\ &= \max_{h_t} x_t' \left(P + Q F_\iota + F_\iota' H_\iota F_\iota\right) x_t + x_t' \left(R + F_\iota' H_{\iota h}\right) h_t + H_h h_t^2 + \gamma E\left[V^n(x_{t+1})\right] \end{aligned} \tag{A5}$$

  subject to the law of motion (15), with ι_t replaced by $F_\iota x_t$:

$$x_{t+1} = \left(A + B F_\iota\right) x_t + C h_t + D \mu_{i,t+1} \tag{A6}$$

With the linear-quadratic setting, we know that the value functions V^k and V^n takes the form

$$V^k(x_t) = x_t' \hat{P}_k x_t + \hat{Q}_k, V^n(x_t) = x_t' \hat{P}_n x_t + \hat{Q}_n$$

Plug it into the Bellman equations: capital manager's problem becomes

$$
\begin{aligned}
& x_t' \left[P + RF_h + F_h' H_h F_h + \gamma(A + CF_h)'\hat{P}_k(A + CF_h)\right] x_t + i_t'(H_i + \gamma B'\hat{P}_k B)i_t \\
& + i_t'(H_{ih}F_h + Q' + \gamma B'\hat{P}_k(A + CF_h))x_t + \gamma x_t'(A + CF_h)'\hat{P}_k B i_t \\
& = \cdots + i_t' \left[H_{ih}F_h + Q' + \gamma B'(\hat{P}_k + \hat{P}_k')(A + CF_h)\right] x_t
\end{aligned}
$$

take first order condition on ι:

$$
2(H_i + \gamma B'\hat{P}_k B)\iota_t = -\left[H_{ih}F_h + Q' + \gamma B'(\hat{P}_k + \hat{P}_k')(A + CF_h)\right] x_t
$$

we get the policy matrix F_ι that solves (A3):

$$
F_\iota = -\frac{1}{2}\left(H_\iota + \gamma B'\hat{P}_k B\right)^{-1}\left[Q' + H_{\iota h}F_h + \gamma B'(\hat{P}_k + \hat{P}_k')(A + CF_h)\right] \tag{A7}
$$

where $\hat{P}_k$ is defined by Riccati equation

$$
\begin{aligned}
\hat{P}_k = {} & P + RF_h + F_h' H_h F_h + \gamma(A + CF_h)'\hat{P}_k(A + CF_h) \\
& - \frac{1}{4}\left[Q' + H_{\iota h}F_h + 2\gamma B'\hat{P}_k(A + CF_h)\right]'\left(H_\iota + \gamma B'\hat{P}_k B\right)^{-1}\left[Q' + H_{\iota h}F_h + 2\gamma B'\hat{P}_k(A + CF_h)\right]
\end{aligned} \tag{A8}
$$

and similarly the solution to (A5) is

$$
F_h = -\frac{1}{2}\left(H_h + \gamma C'\hat{P}_n C\right)^{-1}\left[R' + H_{\iota h}F_\iota + 2\gamma C'\hat{P}_n(A + BF_\iota)\right] \tag{A9}
$$

where $\hat{P}_n$ is defined by Riccati equation

$$
\begin{aligned}
\hat{P}_n = {} & P + QF_\iota + F_\iota' H_\iota F_\iota + (A + BF_\iota)'\hat{P}_n(A + BF_\iota) \\
& - \frac{1}{4}\left[R' + H_{\iota h}F_\iota + 2\gamma C'\hat{P}_n(A + BF_\iota)\right]'\left(H_h + \gamma C'\hat{P}_n C\right)^{-1}\left[R' + H_{\iota h}F_\iota + 2\gamma C'\hat{P}_n(A + BF_\iota)\right]
\end{aligned} \tag{A10}
$$

A.3.2 Part II: linearity of MPBE

Now that we have characterized beliefs given policy F_ι, F_h, I go back to the Bellman equations and derive these policy matrices F_ι, F_h. Under incomplete information, the capital manager's Bellman equation is

$$V^k(x_t) = \max_{\iota_t} E\left[x_t'\left(P + RF_h + F_h'H_hF_h\right)x_t + x_t'\left(Q + F_h'H_{\iota h}\right)\iota_t + H_\iota\iota_t^2 + \gamma V^k(x_{t+1})\middle|\mathcal{I}_t^k\right] \tag{A11}$$

To save some notation, let $\tilde{P}^k \equiv P + RF_h + F_h'H_hF_h$ and $\tilde{Q}^k \equiv Q + F_h'H_{\iota h}$.

From the previous section, we know that $x_t = [s_t, a_t]'$ where $s_t \equiv [1, k_{i,t-1}, n_{i,t-1}]'$ is the perfectly observable part, and a_t is the partially observed productivity. We know that the time-t estimate of x_t for the capital manager is

$$\hat{x}_t^k \equiv [s_t, \hat{a}_t^k]'$$

To see how the noisy information changes the Bellman equation, we first look at the first term $x_t'\left(P + RF_h + F_h'H_hF_h\right)x_t$. Put it in the conditional expectation, we get

$$\begin{aligned}
E\left[x_t'\left(P + RF_h + F_h'H_hF_h\right)x_t\middle|\mathcal{I}_t^k\right] &= E\left[x_t'\tilde{P}^kx_t\middle|\mathcal{I}_t^k\right] \\
&= E\left[[s_t', a_t']\begin{bmatrix}\tilde{P}_{11}^k & \tilde{P}_{12}^k \\ \tilde{P}_{21}^k & \tilde{P}_{22}^k\end{bmatrix}\begin{bmatrix}s_t \\ a_t\end{bmatrix}\middle|\mathcal{I}_t^k\right] \\
&= E\left[s_t'\tilde{P}_{11}^ks_t + s_t'\tilde{P}_{12}^ka_t + a_t'\tilde{P}_{21}^ks_t + a_t'\tilde{P}_{22}^ka_t\middle|\mathcal{I}_t^k\right] \\
&= s_t'\tilde{P}_{11}^ks_t + s_t'\tilde{P}_{12}^k\hat{a}_t^k + (\hat{a}_t^k)'\tilde{P}_{21}^ks_t + E\left[a_t'\tilde{P}_{22}^ka_t\middle|\mathcal{I}_t^k\right]
\end{aligned}$$

We know that

$$\begin{aligned}
E\left[a_t'\tilde{P}_{22}^ka_t\middle|\mathcal{I}_t^k\right] &= E\left[tr\left(a_t'\tilde{P}_{22}^ka_t\right)\middle|\mathcal{I}_t^k\right] \\
&= E\left[tr\left(\tilde{P}_{22}^ka_ta_t'\right)\middle|\mathcal{I}_t^k\right] = tr\left(\tilde{P}_{22}^kE\left[a_ta_t'\middle|\mathcal{I}_t^k\right]\right) \\
&= tr\left(\tilde{P}_{22}^k\left[\hat{a}_t^k(\hat{a}_t^k)' + \hat{\Sigma}^k\right]\right) = tr\left(\tilde{P}_{22}^k\hat{\Sigma}^k\right) + tr\left(\tilde{P}_{22}^k\hat{a}_t^k(\hat{a}_t^k)'\right) \\
&= tr\left(\tilde{P}_{22}^k\hat{\Sigma}^k\right) + (\hat{a}_t^k)'\tilde{P}_{22}^k\hat{a}_t^k
\end{aligned}$$

Plug it back:

$$\begin{aligned}
E\left[x_t'\left(P + RF_h + F_h'H_hF_h\right)x_t\middle|\mathcal{I}_t^k\right] &= s_t'\tilde{P}_{11}^ks_t + s_t'\tilde{P}_{12}^k\hat{a}_t^k + (\hat{a}_t^k)'\tilde{P}_{21}^ks_t + E\left[a_t'\tilde{P}_{22}^ka_t\middle|\mathcal{I}_t^k\right] \\
&= s_t'\tilde{P}_{11}^ks_t + s_t'\tilde{P}_{12}^k\hat{a}_t^k + (\hat{a}_t^k)'\tilde{P}_{21}^ks_t + tr\left(\tilde{P}_{22}^k\hat{\Sigma}^k\right) + (\hat{a}_t^k)'\tilde{P}_{22}^k\hat{a}_t^k
\end{aligned}$$

$$
\begin{aligned}
&= tr\left(\tilde{P}_{22}^{k}\hat{\Sigma}^{k}\right) + [s_t', (\hat{a}_t^k)'] \begin{bmatrix} \tilde{P}_{11}^{k} & \tilde{P}_{12}^{k} \\ \tilde{P}_{21}^{k} & \tilde{P}_{22}^{k} \end{bmatrix} \begin{bmatrix} s_t \\ \hat{a}_t^k \end{bmatrix} \\
&= tr\left(\tilde{P}_{22}^{k}\hat{\Sigma}^{k}\right) + (\hat{x}_t^k)'\tilde{P}^k\hat{x}_t^k
\end{aligned}
$$

The rest of terms in the quadratic profit function is straight forward. For the second term,

$$
E\left[x_t'\left(Q + F_h'H_{\iota h}\right)\iota_t \middle| \mathcal{I}_t^k\right] = (\hat{x}_t^k)'\tilde{Q}^k\iota_t
$$

and the third term can be taken out of the expectation directly since ι_t is a function of elements in $\mathcal{I}_t^k$.

As a result, we have

$$
\begin{aligned}
V^k(x_t) = \max_{\iota_t} tr\left(\tilde{P}_{22}^{k}\hat{\Sigma}^{k}\right) &+ (\hat{x}_t^k)'\left(P + RF_h + F_h'H_hF_h\right)\hat{x}_t^k + (\hat{x}_t^k)'\left(Q + F_h'H_{\iota h}\right)\iota_t + H_\iota\iota_t^2 \\
&+ \gamma E\left[V^k(x_{t+1}) \middle| \mathcal{I}_t^k\right] \qquad \text{(A12)}
\end{aligned}
$$

Compared with the complete-information case, we can see that the expected profit under incomplete information is different in the following aspects:

- replacing the true state variable x_t by the belief $\hat{x}_t^k$;
- adding a constant term that contains the posterior uncertainty $\hat{\Sigma}^k$.

Notably, the posterior variance $\hat{\Sigma}^k$ enters the value function only through the constant trace term and does not affect the coefficients on $\hat{x}_t^k$ or ι_t. Hence it does not affect the first-order condition, as implied by certainty equivalence in this linear-quadratic problem.

Again, I guess that the value function takes the form

$$
(\hat{x}_t^k)'\hat{P}_k\hat{x}_t^k + \hat{Q}_k
$$

so that

$$
\begin{aligned}
E\left[V^k(x_{t+1}) \middle| \mathcal{I}_t^k\right] &= E\left[(\hat{x}_{t+1}^k)'\hat{P}_k\hat{x}_{t+1}^k + \hat{Q}_k \middle| \mathcal{I}_t^k\right] \\
&= tr\left(\hat{P}_k\Xi_{t+1|t}^k\right) + \mathbb{E}_t^k[\hat{x}_{t+1}^k]'\hat{P}_k\mathbb{E}_t^k[\hat{x}_{t+1}^k] + \hat{Q}_k,
\end{aligned}
$$

where $\Xi^k_{t+1|t} \equiv \mathrm{Var}(\hat{x}^k_{t+1} \mid \mathcal{I}^k_t)$. By Lemma 2, this conditional variance is determined by the stationary signal variances and does not depend on ι_t. Substituting into (A12) and taking the first-order condition therefore gives the same F_ι as in (A7), the full-information benchmark. The coefficient matrix $\hat{P}_k$ is pinned down by the same Riccati equation (A8); only the value-function constant $\hat{Q}_k$ changes to absorb the trace terms.

Similarly, for the labor manager, the policy matrix F_h will be exactly the same as the (A9), with Riccatti equation defined by (A10).

A.4 Proof of Lemma 3

Lemma 1 establishes that, conditional on the managers' Gaussian beliefs, the certainty-equivalent best responses are the linear policies (16) and (17). When $F^a_\iota F^a_h \neq 0$, Lemma 2 establishes that these policies reveal the two posterior means at the end of each period and therefore induce the common-prior recursion (23). Given that prior, equations (19)–(22) and the conditional-variance formulas in Lemma 2 are exactly the beliefs generated by Bayes' rule.

Thus the policies are optimal given the beliefs, and the beliefs are those generated by the policies. These are the two consistency conditions in the definition of a linear Markov Perfect Bayesian Equilibrium, which proves the result. In particular, the scalar signal-revelation argument means that the equilibrium does not require the pre-estimate or recursive forecast-error covariance objects used in the earlier filtering formulation.

A.5 Analytical Computation of the Stationary Distribution and Model Moments

For a candidate parameter vector Θ, the linear-Gaussian structure of the model allows the population moments to be computed without simulating a finite panel. The computation proceeds as follows.

1. Solve for the policy rules F_ι and F_h. Given these rules, solve the scalar fixed-point equation in Lemma 2 for $\bar{\Sigma}$ and obtain the posterior variances and belief-updating coefficients from the closed-form expressions in that lemma.

2. Stack capital, labor, the fundamental, and the posterior and higher-order belief means needed by the policy rules in a state vector z_{it}. The equilibrium law of motion can be written as

$$z_{it} = \mathcal{C}(\Theta) z_{i,t-1} + \mathcal{D}(\Theta) u_{it} + \mathcal{D}_\delta(\Theta)\delta_i, \tag{A13}$$

where $u_{it} = (\epsilon^p_{it}, \epsilon^k_{it}, \epsilon^n_{it}, \mu_{it}, \eta^k_{it}, \eta^y_{it})'$ collects the signal errors, the productivity innovation, and the transitory distortions, while $\delta_i = (\delta^k_i, \delta^y_i)'$ collects the permanent distortions.

3. Let $\Sigma_u \equiv \mathrm{var}(u_{it})$. When $\mathcal{C}(\Theta)$ is stable, the covariance matrix $\Omega(\Theta)$ of the dynamic component is the unique solution to the discrete Lyapunov equation

$$\Omega(\Theta) = \mathcal{C}(\Theta)\Omega(\Theta)\mathcal{C}(\Theta)' + \mathcal{D}(\Theta)\Sigma_u\mathcal{D}(\Theta)'. \tag{A14}$$

The permanent distortions shift a firm's stationary mean by $L_\delta(\Theta)\delta_i$, where

$$L_\delta(\Theta) = [I - \mathcal{C}(\Theta)]^{-1}\, \mathcal{D}_\delta(\Theta).$$

Thus, letting $\Sigma_\delta \equiv \mathrm{var}(\delta_i)$, the unconditional cross-sectional covariance matrix is

$$\Sigma_z(\Theta) = \Omega(\Theta) + L_\delta(\Theta)\Sigma_\delta L_\delta(\Theta)'. \tag{A15}$$

4. Construct the contemporaneous and lagged covariances of input levels, input adjustments, beliefs, productivity, and marginal revenue products from equations (A13)–(A15). The model variances and correlations are direct functions of these covariance matrices. The forecast-error coefficients β_h and β_ι are computed as population covariance-to-variance ratios. This produces the complete analytical moment vector $\mathbf{m}(\Theta)$ used in the calibration.

The calibrated parameter vector minimizes the unweighted quadratic distance

$$\hat{\Theta} = \arg\min_{\Theta} \left[\mathbf{m}(\Theta) - \mathbf{m}^{\text{data}}\right]' \left[\mathbf{m}(\Theta) - \mathbf{m}^{\text{data}}\right].$$

Because $\mathbf{m}(\Theta)$ is obtained from population covariance matrices rather than a simulated panel, the criterion contains no model-simulation error.

A.6 Proof of Proposition 1

Define the weights that the capital manager's posterior in Lemma 2 assigns to the common prior, the public signal, and the capital manager's private signal as

$$\omega_0^k \equiv \frac{\hat{\Sigma}^k}{\rho^2\bar{\Sigma} + \sigma_\mu^2}, \qquad \omega_p^k \equiv \frac{\hat{\Sigma}^k}{\sigma_{\epsilon,p}^2}, \qquad \omega_k^k \equiv \frac{\hat{\Sigma}^k}{\sigma_{\epsilon,k}^2}. \tag{A16}$$

The definition of $\hat{\Sigma}^k$ in Lemma 2 implies

$$\omega_0^k + \omega_p^k + \omega_k^k = 1. \tag{A17}$$

We first express the period-t common prior in terms of the surprise in period-$(t-1)$ hiring. Lemma 1, equation (18), and the second-order belief (21) give

$$\begin{aligned} h_{t-1} - \hat{h}_{t-1}^k &= F_h^a \left(\hat{a}_{t-1}^n - \hat{a}_{t-1}^{(k,n)}\right) \\ &= F_h^a \hat{\Sigma}^n \sigma_{\epsilon,n}^{-2} \left(s_{t-1}^n - \hat{a}_{t-1}^k\right). \end{aligned} \tag{A18}$$

Using $(\hat{\Sigma}^k)^{-1} = \bar{\Sigma}^{-1} - \sigma_{\epsilon,n}^{-2}$, equation (23) can therefore be written as

$$\begin{aligned} \bar{a}_t &= \rho\bar{\Sigma}\left[(\hat{\Sigma}^k)^{-1}\hat{a}_{t-1}^k + \sigma_{\epsilon,n}^{-2} s_{t-1}^n\right] \\ &= \rho\left[\hat{a}_{t-1}^k + \frac{\bar{\Sigma}}{F_h^a\hat{\Sigma}^n}\left(h_{t-1} - \hat{h}_{t-1}^k\right)\right]. \end{aligned} \tag{A19}$$

The capital manager's posterior mean in equation (19) is

$$\hat{a}_t^k = \omega_0^k \bar{a}_t + \omega_p^k s_t^p + \omega_k^k s_t^k. \tag{A20}$$

Since $s_t^p = \rho a_{t-1} + \mu_t + \epsilon_t^p$ and $s_t^k = \rho a_{t-1} + \mu_t + \epsilon_t^k$, substituting (A19) into (A20) and using (A17) yields

$$\begin{aligned} \hat{a}_t^k - \rho\hat{a}_{t-1}^k = {} & \rho(\omega_p^k + \omega_k^k)(a_{t-1} - \hat{a}_{t-1}^k) \\ & + \rho\omega_0^k \frac{\bar{\Sigma}}{F_h^a\hat{\Sigma}^n}(h_{t-1} - \hat{h}_{t-1}^k) \\ & + \omega_p^k(\mu_t + \epsilon_t^p) + \omega_k^k(\mu_t + \epsilon_t^k). \end{aligned} \tag{A21}$$

Finally, Lemma 1 gives

$$\begin{aligned}\iota_t - \mathbb{E}^k_{t-1}[\iota_t] &= F^n_\iota \left(n_{t-1} - \mathbb{E}^k_{t-1}[n_{t-1}]\right) + F^a_\iota \left(\hat{a}^k_t - \mathbb{E}^k_{t-1}[a_t]\right) \\ &= F^n_\iota (h_{t-1} - \hat{h}^k_{t-1}) + F^a_\iota (\hat{a}^k_t - \rho \hat{a}^k_{t-1}), \end{aligned} \tag{A22}$$

where the second equality uses $n_{t-1} = n_{t-2} + h_{t-1}$, $n_{t-2} \in \mathcal{I}^k_{t-1}$, and $\mathbb{E}^k_{t-1}[a_t] = \rho \hat{a}^k_{t-1}$. Substituting (A21) into (A22) proves (29), with

$$\beta_1 \equiv \rho F^a_\iota (\omega^k_p + \omega^k_k) = \rho F^a_\iota \hat{\Sigma}^k \left(\sigma^{-2}_{\epsilon,p} + \sigma^{-2}_{\epsilon,k}\right), \tag{A23}$$

$$\begin{aligned}\beta_2 &\equiv F^n_\iota + \rho F^a_\iota \omega^k_0 \frac{\bar{\Sigma}}{F^a_h \hat{\Sigma}^n} \\ &= F^n_\iota + \rho F^a_\iota \frac{\hat{\Sigma}^k \bar{\Sigma}}{(\rho^2 \bar{\Sigma} + \sigma^2_\mu) F^a_h \hat{\Sigma}^n}, \end{aligned} \tag{A24}$$

$$\beta_3 \equiv F^a_\iota \begin{bmatrix} \omega^k_p & \omega^k_k \end{bmatrix} = F^a_\iota \hat{\Sigma}^k \begin{bmatrix} \sigma^{-2}_{\epsilon,p} & \sigma^{-2}_{\epsilon,k} \end{bmatrix}. \tag{A25}$$

Here $H = [1,1]'$ and $w^k_t = [\epsilon^p_t, \epsilon^k_t]'$, so that $\beta_3(H\mu_t + w^k_t)$ equals the last line of (A21) multiplied by F^a_ι.

A.7 Full Statement and Proof of Proposition 2

We suppress the firm subscript and center all variables to simplify notation.

Full statement of Proposition 2 Suppose that the policy rule in Lemma 1 satisfies

$$1 + F^k_\iota \geq 0, \quad F^n_\iota \geq 0, \quad F^k_h \geq 0, \quad 1 + F^n_h \geq 0, \quad F^a_\iota > 0, \quad F^a_h > 0,$$

and the spectral radius of matrix $L \equiv \begin{bmatrix} F^k_\iota + 1 & F^n_\iota \\ F^k_h & F^n_h + 1 \end{bmatrix}$ is below 1.

Define $G_0(\hat{\rho}) \equiv \frac{2\mathcal{C}}{\mathcal{B} + \sqrt{\mathcal{B}^2 + 4\mathcal{A}\mathcal{C}}}$, where $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are defined as

$$\mathcal{A} \equiv \sigma^2_{\epsilon,p} \rho \hat{\rho}^2 (1+\rho), \mathcal{B} \equiv \hat{\rho}(1+\rho)\left[\sigma^2_{\epsilon,p}(1-\rho\hat{\rho}) + \sigma^2_\mu\right], \mathcal{C} \equiv \sigma^2_\mu \rho (1+\hat{\rho})$$

Then the following conclusions follow:

- If $\hat{\rho} > \rho$ and $\underline{G} < G \leq 1$ where $\underline{G} \equiv \max\left\{G_0(\hat{\rho}), 1 - \frac{\rho}{\hat{\rho}}\right\}$, we have $\beta_\iota < 0$ and $\beta_h < 0$;
- If $\hat{\rho} < \rho$ and $0 \leq G < G_0(\hat{\rho})$, then $\beta_\iota > 0$ and $\beta_h > 0$;
- If $\hat{\rho} = \rho$, then $sgn(\beta_\iota) = sgn(\beta_h) = sgn(G^* - G)$ where G^* is the steady-state Bayes-rational Kalman gain.

Proof. Define

$$\phi \equiv (1 - G)\hat{\rho}$$

and let $m_t \equiv \mathbb{F}_t^k[a_t]$. We have

$$m_t = \phi m_{t-1} + G a_t + G\epsilon_t^p. \tag{A26}$$

The manager's subjective conditional forecast is $\mathbb{F}_{t-1}[m_t] = \hat{\rho} m_{t-1}$. Because k_{t-1} and n_{t-1} are known at $t-1$ under common information, the policy function in Lemma 1 therefore implies

$$\iota_t - \mathbb{F}_{t-1}[\iota_t] = F_\iota^a \delta_t, \qquad \delta_t \equiv m_t - \hat{\rho} m_{t-1}. \tag{A27}$$

Let $x_t \equiv [k_t, n_t]'$, $d_t \equiv [\iota_t, h_t]'$ and $f \equiv [F_\iota^a, F_h^a]'$. Since $x_t = x_{t-1} + d_t$, the policy rules can be written as $x_t = L x_{t-1} + f m_t$. We have an MA-representation of x_t and d_{t-1}:

$$x_t = \sum_{j=0}^{\infty} L^j f m_{t-j}, \qquad d_{t-1} = \sum_{j=0}^{\infty} L^j f \Delta m_{t-1-j}. \tag{A28}$$

Now, let's consider the signs of β_ι and β_h. We have

$$\begin{bmatrix} \beta_\iota \\ \beta_h \end{bmatrix} = F_\iota^a \begin{bmatrix} \mathrm{Var}(\iota_{t-1})^{-1} \\ \mathrm{Var}(h_{t-1})^{-1} \end{bmatrix} \odot \mathrm{Cov}(\iota_t - \mathbb{F}_{t-1}[\iota_t], d_{t-1}) = \begin{bmatrix} \mathrm{Var}(\iota_{t-1})^{-1} \\ \mathrm{Var}(h_{t-1})^{-1} \end{bmatrix} \odot \sum_{j=0}^{\infty} F_\iota^a L^j f \underbrace{\mathrm{Cov}(\delta_t, \Delta m_{t-1-j})}_{\equiv g_j}$$

With our assumptions on the policy coefficients, $L^j f \geq 0$ for all j. Thus it remains to determine the sign of g_j for $j \geq 0$. The rest of the proof characterizes the sign of g_j.

Let

$$V_a \equiv \mathrm{Var}(a_t) = \frac{\sigma_\mu^2}{1 - \rho^2}, \qquad c_{am} \equiv \mathrm{Cov}(a_t, m_t), \qquad V_m \equiv \mathrm{Var}(m_t).$$

Equation (A26) implies that

$$c_{am} = \frac{GV_a}{1-\rho\phi}, \tag{A29}$$

$$V_m = \frac{G^2}{1-\phi^2}\left[\sigma^2_{\epsilon,p} + V_a\frac{1+\rho\phi}{1-\rho\phi}\right]. \tag{A30}$$

Let $\gamma_j \equiv \mathrm{Cov}(m_t, m_{t-j})$. For $j \geq 1$,

$$\gamma_j = \phi\gamma_{j-1} + G\rho^j c_{am}.$$

Whenever $\phi < \rho$, its solution is

$$\gamma_j = K_\rho\rho^j + K_\phi\phi^j, \qquad K_\rho \equiv \frac{G\rho c_{am}}{\rho-\phi}, \qquad K_\phi \equiv V_m - K_\rho. \tag{A31}$$

This expression also holds at $j = 0$. Consequently,

$$g_j = [\gamma_{j+1} - \hat{\rho}\gamma_j] - [\gamma_{j+2} - \hat{\rho}\gamma_{j+1}] = B_\rho\rho^j + B_\phi\phi^j, \tag{A32}$$
$$\text{where } B_\rho \equiv K_\rho(\rho-\hat{\rho})(1-\rho), \qquad B_\phi \equiv K_\phi(\phi-\hat{\rho})(1-\phi)$$

We first pin down the sign of g_0. Substituting (A29) and (A30) into (A32) at $j = 0$ gives

$$g_0 = -\frac{G^2}{(1+\rho)(1+\phi)(1-\rho\phi)}\mathcal{D}(G), \tag{A33}$$

where

$$\begin{aligned}\mathcal{D}(G) &\equiv G\sigma^2_{\epsilon,p}\hat{\rho}(1+\rho)(1-\rho\phi) + \sigma^2_\mu\left[G\hat{\rho}(1+\rho) - \rho(1+\hat{\rho})\right] \\ &= \mathcal{A}G^2 + \mathcal{B}G - \mathcal{C}\end{aligned} \tag{A34}$$

and $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are defined in the full statement of the proposition. Because $\mathcal{A}, \mathcal{B}, \mathcal{C} > 0$, $\mathcal{D}(G)$ is strictly increasing for $G > 0$, starts below zero, and has the unique positive root $G_0(\hat{\rho})$. It follows that

$$g_0 > 0 \iff G < G_0(\hat{\rho}), \qquad g_0 < 0 \iff G > G_0(\hat{\rho}). \tag{A35}$$

Next, we pin down the sign g_j. Under $\phi < \rho$, $K_\rho > 0$, so B_ρ has the sign of $\rho - \hat{\rho}$. Moreover,

$$\frac{g_j}{\rho^j} = B_\rho + B_\phi \left(\frac{\phi}{\rho}\right)^j .$$

Hence we have the following:

- If $\rho < \hat{\rho}$, then $B_\rho < 0$. It turns out that $g_j \leq 0$ for all $j \geq 0$ as long as $G \geq \underline{G}$. The proof is as follows:
  - When $G \geq \underline{G} \geq G_0(\hat{\rho})$, then $g_0 < 0$ from (A35);
  - If $B_\phi \leq 0$ as well, then $g_j < 0$ for all j.
  - If $B_\phi > 0$, since $\phi < \rho$, as j goes up, $(\phi/\rho)^j$ goes down, and therefore $\frac{g_j}{\rho^j} \leq g_0 < 0$ for $j > 0$.
- If $\rho > \hat{\rho}$, then $B_\rho > 0$. It turns out that $g_j \geq 0$ for all $j \geq 0$ as long as $G \leq G_0(\hat{\rho})$:
  - When $G \leq G_0(\hat{\rho})$, then $g_0 \geq 0$ from (A35);
  - If $B_\phi \geq 0$ as well, then $g_j > 0$ for all j.
  - If $B_\phi < 0$, since $\phi < \rho$, as j goes up, $(\phi/\rho)^j$ goes down, and therefore $\frac{g_j}{\rho^j} \geq g_0 > 0$ for $j > 0$.

It remains to prove the case $\hat{\rho} = \rho$. In this case, $B_\rho = 0$, so all g_j have the sign of g_0. In addition,

$$\frac{\mathcal{D}(G)}{\rho(1+\rho)} = \sigma_{\epsilon,p}^2 \rho^2 G^2 + \left[\sigma_{\epsilon,p}^2 (1-\rho^2) + \sigma_\mu^2\right] G - \sigma_\mu^2. \tag{A36}$$

The unique positive root of the right-hand side is precisely the steady-state Bayes-rational Kalman gain G^*. This follows from the standard scalar Riccati equations

$$V^- = \rho^2 V + \sigma_\mu^2, \qquad G^* = \frac{V^-}{V^- + \sigma_{\epsilon,p}^2}, \qquad V = (1 - G^*)V^-.$$

Equation (A33) then implies that all g_j, and hence both projection coefficients, are positive below G^*, zero at G^*, and negative above G^*. This completes the proof.

A.8 Proof of Corollary 1

To simplify notation, define the capital manager's period-$(t-1)$ productivity nowcast error and hiring forecast error as

$$e^k_{t-1} \equiv a_{t-1} - \hat{a}^k_{t-1}, \qquad q^h_{t-1} \equiv h_{t-1} - \hat{h}^k_{t-1}.$$

Proposition 1 implies

$$\iota_t - \mathbb{E}^k_{t-1}[\iota_t] = \beta_1 e^k_{t-1} + \beta_2 q^h_{t-1} + \eta_t, \tag{A37}$$

where $\eta_t \equiv \beta_3(H\mu_t + w^k_t)$ is independent of period-$(t-1)$ hiring. Hence

$$\operatorname{Cov}\left(\iota_t - \mathbb{E}^k_{t-1}[\iota_t], h_{t-1}\right) = \beta_1 \operatorname{Cov}(e^k_{t-1}, h_{t-1}) + \beta_2 \operatorname{Cov}(q^h_{t-1}, h_{t-1}). \tag{A38}$$

Because $\hat{h}^k_{t-1} = \mathbb{E}^k_{t-1}[h_{t-1}]$, the hiring forecast error is orthogonal to the capital manager's information set. It follows that

$$\operatorname{Cov}(q^h_{t-1}, h_{t-1}) = \operatorname{Var}(q^h_{t-1}) > 0, \tag{A39}$$

where strict positivity follows from the segmentation assumption in the corollary.

Similarly, $\mathbb{E}^k_{t-1}[e^k_{t-1}] = 0$, so that e^k_{t-1} is orthogonal to $\hat{h}^k_{t-1}$ and

$$\operatorname{Cov}(e^k_{t-1}, h_{t-1}) = \operatorname{Cov}(e^k_{t-1}, q^h_{t-1}).$$

Equation (A18) gives

$$\begin{aligned}\operatorname{Cov}(e^k_{t-1}, q^h_{t-1}) &= F^a_h \hat{\Sigma}^n \sigma^{-2}_{\epsilon,n} \operatorname{Cov}\left(a_{t-1} - \hat{a}^k_{t-1}, s^n_{t-1} - \hat{a}^k_{t-1}\right) \\ &= F^a_h \hat{\Sigma}^n \sigma^{-2}_{\epsilon,n} \hat{\Sigma}^k > 0.\end{aligned} \tag{A40}$$

The second equality uses $s^n_{t-1} = a_{t-1} + \epsilon^n_{t-1}$, the independence of ϵ^n_{t-1} from the capital manager's nowcast error, and $\operatorname{Var}(a_{t-1} - \hat{a}^k_{t-1}) = \hat{\Sigma}^k$.

Finally, equations (A23) and (A24) imply

$$\begin{aligned}\beta_1 &= \rho F^a_\iota \hat{\Sigma}^k \left(\sigma^{-2}_{\epsilon,p} + \sigma^{-2}_{\epsilon,k}\right) \geq 0, \\ \beta_2 &= F^n_\iota + \rho F^a_\iota \frac{\hat{\Sigma}^k \bar{\Sigma}}{(\rho^2 \bar{\Sigma} + \sigma^2_\mu) F^a_h \hat{\Sigma}^n} > 0.\end{aligned}$$

Substituting these signs together with (A39) and (A40) into (A38) gives

$$\operatorname{Cov}\left(\iota_t - \mathbb{E}^k_{t-1}[\iota_t], h_{t-1}\right) > 0.$$

Since $\operatorname{Var}(h_{t-1}) > 0$, dividing by that variance proves $\beta_h^{\text{seg}} = \beta_h^R > 0$.

A.9 Full Statement and Proof of Proposition 3

We suppress the firm subscript and center all log variables around their frictionless steady-state values. Throughout this subsection, σ denotes the capital–labor elasticity of substitution rather than the standard deviation of a shock.

Full statement of Proposition 3. Suppose the firm chooses capital and labor flexibly at the fixed factor prices $\bar{R}$ and $\bar{W}$, so that $\xi_k = \xi_n = 0$, and its gross revenue is given by (32). For $j \in \{k, n\}$, define the log deviation of factor-augmenting technology as $a_{j,t} \equiv \log(A_{j,t}/\bar{A}_j)$ and let it follow

$$a_{j,t} = \rho a_{j,t-1} + u_{j,t}, \qquad 0 \le \rho < 1, \tag{A41}$$

and let the firm observe

$$s_{j,t} = a_{j,t} + \varepsilon_{j,t}. \tag{A42}$$

The innovations $u_{k,t}$ and $u_{n,t}$ and the signal errors $\varepsilon_{k,t}$ and $\varepsilon_{n,t}$ are mutually independent across factors and time, with strictly positive variances. The firm has common information and correctly perceives all innovation and signal-noise variances, but perceives the persistence of both technologies to be $\widehat{\rho}$, where

$$0 \le \rho < \widehat{\rho} < 1. \tag{A43}$$

It uses the corresponding stationary Kalman filters. In particular, define

$$\widehat{a}_{j,t} \equiv \mathbb{F}_t[a_{j,t}] = \widehat{\rho}\,\widehat{a}_{j,t-1} + G_j\left(s_{j,t} - \widehat{\rho}\,\widehat{a}_{j,t-1}\right), \qquad G_j \in (0,1). \tag{A44}$$

Here the subscript on $\widehat{a}_{j,t}$ identifies the factor-specific technology, not the manager forming the belief.

Let $\bar{s}_K$ be the steady-state effective-capital share in (33) and define

$$\psi_{kk} \equiv \bar{s}_K(\epsilon - 1) + (1 - \bar{s}_K)(\sigma - 1), \qquad \psi_{kn} \equiv (1 - \bar{s}_K)(\epsilon - \sigma), \tag{A45}$$

$$\psi_{nk} \equiv \bar{s}_K(\epsilon - \sigma), \qquad \psi_{nn} \equiv (1 - \bar{s}_K)(\epsilon - 1) + \bar{s}_K(\sigma - 1). \tag{A46}$$

Suppose that $\epsilon > 1$, $\sigma > 0$, and

$$\psi_{kk} > 0, \qquad \psi_{nn} > 0. \tag{A47}$$

The Cobb–Douglas case $\sigma = 1$ is interpreted by continuity. Define flexible log factor choices k_t^* and n_t^*, investment and hiring by

$$\iota_t = k_t^* - k_{t-1}^*, \qquad h_t = n_t^* - n_{t-1}^*, \tag{A48}$$

and the investment forecast error by

$$FE_t^\iota \equiv \iota_t - \mathbb{F}_{t-1}[\iota_t]. \tag{A49}$$

Provided that the two regressor variances are positive, define the population projection coefficients

$$\beta_\iota \equiv \frac{\mathrm{Cov}(FE_t^\iota, \iota_{t-1})}{\mathrm{Var}(\iota_{t-1})}, \qquad \beta_h \equiv \frac{\mathrm{Cov}(FE_t^\iota, h_{t-1})}{\mathrm{Var}(h_{t-1})}. \tag{A50}$$

Then

$$\beta_\iota < 0, \qquad \mathrm{sgn}(\beta_h) = \mathrm{sgn}(\sigma - \epsilon). \tag{A51}$$

Consequently,

$$\beta_\iota < 0 < \beta_h \quad \Longleftrightarrow \quad \sigma > \epsilon. \tag{A52}$$

Condition (A47) is automatic when $\sigma \geq 1$. When $0 < \sigma < 1$, it is equivalent to

$$\frac{1 - \sigma}{\epsilon - \sigma} < \bar{s}_K < \frac{\epsilon - 1}{\epsilon - \sigma}. \tag{A53}$$

Proof. The proof proceeds in three steps.

Step 1: Log-linear factor demands. Define

$$\vartheta \equiv 1 - \frac{1}{\epsilon}$$

and the time-t effective-capital share

$$s_{K,t} \equiv \frac{\alpha^{1/\sigma}(A_{k,t}K_t)^{1-1/\sigma}}{\alpha^{1/\sigma}(A_{k,t}K_t)^{1-1/\sigma} + (1-\alpha)^{1/\sigma}(A_{n,t}N_t)^{1-1/\sigma}}. \tag{A54}$$

The revenue elasticities with respect to capital and labor are

$$\frac{\partial \log Y_t}{\partial \log K_t} = \vartheta s_{K,t}, \qquad \frac{\partial \log Y_t}{\partial \log N_t} = \vartheta(1 - s_{K,t}). \tag{A55}$$

Therefore the flexible-input first-order conditions are

$$\vartheta s_{K,t}\frac{Y_t}{K_t} = \bar{R}, \qquad \vartheta(1 - s_{K,t})\frac{Y_t}{N_t} = \bar{W}. \tag{A56}$$

Certainty equivalence in the first-order approximation means that the firm evaluates these conditions using $\widehat{a}_{k,t}$ and $\widehat{a}_{n,t}$. Let y_t be the log deviation of revenue and let k_t^* and n_t^* denote the log deviations of the flexible factor choices. Log-linearizing revenue gives

$$y_t = \vartheta\left[\bar{s}_K(\widehat{a}_{k,t} + k_t^*) + (1 - \bar{s}_K)(\widehat{a}_{n,t} + n_t^*)\right]. \tag{A57}$$

The corresponding log-linearizations of the two CES shares are

$$\log\left(\frac{s_{K,t}}{\bar{s}_K}\right) = (1 - \bar{s}_K)\left(1 - \frac{1}{\sigma}\right)\left[(\widehat{a}_{k,t} + k_t^*) - (\widehat{a}_{n,t} + n_t^*)\right], \tag{A58}$$

$$\log\left(\frac{1 - s_{K,t}}{1 - \bar{s}_K}\right) = -\bar{s}_K\left(1 - \frac{1}{\sigma}\right)\left[(\widehat{a}_{k,t} + k_t^*) - (\widehat{a}_{n,t} + n_t^*)\right]. \tag{A59}$$

Since factor prices are fixed, the log-linearized first-order conditions from (A56) are

$$0 = y_t - k_t^* + (1 - \bar{s}_K)\left(1 - \frac{1}{\sigma}\right)\left[(\widehat{a}_{k,t} + k_t^*) - (\widehat{a}_{n,t} + n_t^*)\right], \tag{A60}$$

$$0 = y_t - n_t^* - \bar{s}_K\left(1 - \frac{1}{\sigma}\right)\left[(\widehat{a}_{k,t} + k_t^*) - (\widehat{a}_{n,t} + n_t^*)\right]. \tag{A61}$$

Subtracting (A61) from (A60) and rearranging yields

$$k_t^* - n_t^* = (\sigma - 1)(\widehat{a}_{k,t} - \widehat{a}_{n,t}). \tag{A62}$$

Multiplying (A60) by $\bar{s}_K$, multiplying (A61) by $1 - \bar{s}_K$, and adding cancels the share terms and gives

$$y_t = \bar{s}_K k_t^* + (1 - \bar{s}_K)n_t^*.$$

Combining this equality with (A57) gives

$$\bar{s}_K k_t^* + (1-\bar{s}_K) n_t^* = (\epsilon - 1)\left[\bar{s}_K \widehat{a}_{k,t} + (1-\bar{s}_K)\widehat{a}_{n,t}\right]. \tag{A63}$$

Solving (A62) and (A63) for the two choices gives

$$k_t^* = \psi_{kk}\widehat{a}_{k,t} + \psi_{kn}\widehat{a}_{n,t}, \tag{A64}$$

$$n_t^* = \psi_{nk}\widehat{a}_{k,t} + \psi_{nn}\widehat{a}_{n,t}, \tag{A65}$$

with the coefficients in (A45) and (A46).

For $\sigma \geq 1$, both ψ_{kk} and ψ_{nn} are strictly positive because $\epsilon > 1$ and $\bar{s}_K \in (0,1)$. If $0 < \sigma < 1$, rewriting the two inequalities in (A47) gives exactly (A53).

Step 2: The covariance generated by over-extrapolation. For each $j \in \{k, n\}$, define

$$\delta_{j,t} \equiv \widehat{a}_{j,t} - \widehat{\rho}\,\widehat{a}_{j,t-1}, \qquad d_{j,t-1} \equiv \widehat{a}_{j,t-1} - \widehat{a}_{j,t-2}, \tag{A66}$$

and let

$$\chi_j \equiv \mathrm{Cov}(\delta_{j,t}, d_{j,t-1}). \tag{A67}$$

We now show that $\chi_j < 0$ whenever $\widehat{\rho} > \rho$.

Let $\ell_j \equiv 1 - G_j$ and $\phi_j \equiv \ell_j \widehat{\rho}$. Under the true data-generating process, (A44) can be written as

$$\widehat{a}_{j,t} = \phi_j \widehat{a}_{j,t-1} + G_j a_{j,t} + G_j \varepsilon_{j,t}. \tag{A68}$$

Define

$$V_{a,j} \equiv \mathrm{Var}(a_{j,t}) = \frac{\sigma_{u,j}^2}{1-\rho^2}, \qquad c_j \equiv \mathrm{Cov}(a_{j,t}, \widehat{a}_{j,t}), \qquad V_{\widehat{a},j} \equiv \mathrm{Var}(\widehat{a}_{j,t}).$$

Equation (A68) implies

$$c_j = \frac{G_j V_{a,j}}{1-\rho\phi_j}, \tag{A69}$$

$$V_{\widehat{a},j} = \frac{G_j^2}{1-\phi_j^2}\left[\sigma_{\varepsilon,j}^2 + V_{a,j}\frac{1+\rho\phi_j}{1-\rho\phi_j}\right]. \tag{A70}$$

Let $\gamma_{j,q} \equiv \mathrm{Cov}(\widehat{a}_{j,t}, \widehat{a}_{j,t-q})$. For $q \geq 1$, (A68) gives

$$\gamma_{j,q} = \phi_j \gamma_{j,q-1} + G_j \rho^q c_j. \tag{A71}$$

In particular,

$$\gamma_{j,1} = \phi_j V_{\widehat{a},j} + G_j \rho c_j, \qquad \gamma_{j,2} = \phi_j \gamma_{j,1} + G_j \rho^2 c_j.$$

It follows from (A66) that

$$\begin{aligned}\chi_j &= (1+\widehat{\rho})\gamma_{j,1} - \gamma_{j,2} - \widehat{\rho}\gamma_{j,0} \\ &= (\phi_j - \widehat{\rho})(1-\phi_j)V_{\widehat{a},j} + G_j \rho c_j (1 + \widehat{\rho} - \phi_j - \rho).\end{aligned} \tag{A72}$$

Because G_j is the stationary Kalman gain under the firm's perceived model, the perceived Riccati equations are

$$P_j^- = \widehat{\rho}^2 P_j + \sigma_{u,j}^2, \qquad G_j = \frac{P_j^-}{P_j^- + \sigma_{\varepsilon,j}^2}, \qquad P_j = (1-G_j)P_j^-. \tag{A73}$$

Equivalently,

$$\sigma_{\varepsilon,j}^2 = \frac{\sigma_{u,j}^2 \ell_j}{G_j(1-\widehat{\rho}^2 \ell_j)}. \tag{A74}$$

Substituting (A69), (A70), and (A74) into (A72) and collecting terms yields

$$\chi_j = \frac{\sigma_{u,j}^2 (\rho - \widehat{\rho}) G_j^2 \left[1 - \widehat{\rho}^2 \ell_j + \widehat{\rho}^2 \ell_j^2 (1+\rho)\right]}{(1+\rho)(1+\widehat{\rho}\ell_j)(1-\widehat{\rho}^2\ell_j)(1-\rho\widehat{\rho}\ell_j)}. \tag{A75}$$

Every factor in (A75) other than $\rho - \widehat{\rho}$ is strictly positive. In particular, $1 - \widehat{\rho}^2 \ell_j > 0$ and the bracketed numerator is the sum of this positive term and $\widehat{\rho}^2 \ell_j^2 (1+\rho) > 0$. Consequently,

$$\widehat{\rho} > \rho \quad \Longrightarrow \quad \chi_k < 0, \qquad \chi_n < 0. \tag{A76}$$

Step 3: Signs of the projection coefficients. Under the firm's perceived model,

$$\mathbb{F}_{t-1}[\widehat{a}_{j,t}] = \widehat{\rho}\,\widehat{a}_{j,t-1}.$$

Moreover, k_{t-1}^* is known at time $t-1$. Therefore

$$\begin{aligned}FE_t^{\iota} &= \iota_t - \mathbb{F}_{t-1}[\iota_t] \\ &= k_t^* - \mathbb{F}_{t-1}[k_t^*] \\ &= \psi_{kk}\delta_{k,t} + \psi_{kn}\delta_{n,t}.\end{aligned} \tag{A77}$$

Equations (A64) and (A65) also imply

$$\iota_{t-1} = \psi_{kk} d_{k,t-1} + \psi_{kn} d_{n,t-1}, \tag{A78}$$

$$h_{t-1} = \psi_{nk} d_{k,t-1} + \psi_{nn} d_{n,t-1}. \tag{A79}$$

Let

$$v_j \equiv \mathrm{Var}(d_{j,t-1}) > 0.$$

Independence of the two factor-specific processes implies that all cross-factor covariances in the projection moments are zero. Hence

$$\beta_\iota = \frac{\psi_{kk}^2 \chi_k + \psi_{kn}^2 \chi_n}{\psi_{kk}^2 v_k + \psi_{kn}^2 v_n} < 0, \tag{A80}$$

where the inequality follows from (A76). Similarly,

$$\beta_h = \frac{\psi_{kk}\psi_{nk}\chi_k + \psi_{kn}\psi_{nn}\chi_n}{\psi_{nk}^2 v_k + \psi_{nn}^2 v_n}. \tag{A81}$$

Using (A45) and (A46), the numerator in (A81) factors as

$$\begin{aligned} &\psi_{kk}\psi_{nk}\chi_k + \psi_{kn}\psi_{nn}\chi_n \\ &\qquad = (\epsilon - \sigma)\left[\bar{s}_K \psi_{kk}\chi_k + (1 - \bar{s}_K)\psi_{nn}\chi_n\right]. \end{aligned} \tag{A82}$$

The expression in brackets is strictly negative by (A47) and (A76), while the denominator in (A81) is strictly positive. It follows that

$$\mathrm{sgn}(\beta_h) = -\,\mathrm{sgn}(\epsilon - \sigma) = \mathrm{sgn}(\sigma - \epsilon).$$

Together with (A80), this proves (A51) and (A52).

A.10 Robustness Checks

In this section, I consider several robustness checks to Section 3 and Section 4.

A.10.1 Time-to-build

In the capital adjustment literature, it is common to assume that the capital takes time to build (e.g. Cooper and Haltiwanger (2006), David and Venkateswaran (2019)): today's investment transforms into capital stock with a one-period lag. One may wonder whether time-to-build invalidates the power of empirical test: namely, whether incorporating time-to-build in capital alone makes the coefficient of interest $\tilde{\beta}_h \neq 0$.

I verify that time-to-build in capital does not invalidate the regression test. In our framework of Dynamic Narrow Framing, the only change is on the profit function: without time-to-build, we have

$$\Pi(K, N; A, K_{-1}, N_{-1}) = \underbrace{AK^{\alpha(1-1/\epsilon)}N^{(1-\alpha)(1-1/\epsilon)}}_{=\mathcal{R}(A,K,N)} - RK - WN - \text{adj. costs}$$

With time-to-build on capital, it seems that the only change is on the profit function:

$$\Pi(K, N; A, K_{-1}, N_{-1}) = AK_{-1}{}^{\alpha(1-1/\epsilon)}N^{(1-\alpha)(1-1/\epsilon)} - RK - WN - \text{adj. costs} \tag{A83}$$

In other words, with the time-to-build friction, capital for production today is predetermined, and the cost for tomorrow's capital is paid today. This friction changes the matrices in the log-quadratic profit function (A2): now the matrix $P, Q, R, H_\iota, H_h, H_{\iota h}$ becomes

$$P \equiv \begin{bmatrix} \frac{1}{2\epsilon}\bar{Y} & 0 & 0 & \frac{1}{2}(1-1/\epsilon)\bar{Y} \\ * & \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) & \frac{1}{2}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} & \frac{1}{2}\alpha(1-1/\epsilon)^2\bar{Y} \\ * & * & \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) & \frac{1}{2}(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ * & * & * & \frac{1}{2}(1-1/\epsilon)^2\bar{Y} \end{bmatrix} \tag{A84}$$

$$Q \equiv \begin{bmatrix} -\bar{R}\bar{K} \\ -\bar{R}\bar{K} \\ 0 \\ 0 \end{bmatrix} \text{ and } R \equiv \begin{bmatrix} 0 \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) \\ (1-\alpha)(1-1/\epsilon)^2\bar{Y} \end{bmatrix} \tag{A85}$$

and

$$H_\iota \equiv -\frac{1}{2}(\bar{R}\bar{K} + \xi_k\bar{K}), H_h \equiv -\frac{1}{2}([1-(1-\alpha)(1-1/\epsilon)]\bar{W}\bar{N} + \xi_n\bar{N}) \text{ and } H_{\iota h} = 0 \tag{A86}$$

The learning process of firm managers is not changed by the time-to-build friction, and therefore the intuition we built in Section 3 remains true: lagged hiring cannot forecast the investment forecast error.

A.10.2 Reduced-form Distortions

I show that reduced-form distortions cannot reproduce the sign patterns by themselves. Consider reduced-form distortions as in Hsieh and Klenow (2009): the profit of the firm is given by

$$\Pi_{it} = (1-\tau^y_{it})R(K_{it}, N_{it}; A_{it}) - (1+\tau^k_{it})R_t K_{it} - W_t N_{it} \\ - \frac{\xi_k}{2}\left(\frac{K_{it}}{K_{i,t-1}} - 1\right)^2 K_{i,t-1} - \frac{\xi_n}{2}\left(\frac{N_{it}}{N_{i,t-1}} - 1\right)^2 N_{i,t-1} \quad \text{(A87)}$$

where τ^y_{it} is a revenue distortion and τ^k_{it} is a capital-specific distortion. I follow David and Venkateswaran (2019) and assume that these distortions have the following structure:

$$\tau^y_{it} = \gamma^y a_{it} + \delta^y_i + \eta^y_{it} \quad \text{(A88)}$$

$$\tau^k_{it} = \gamma^k a_{it} + \delta^k_i + \eta^k_{it} \quad \text{(A89)}$$

where $\gamma^y, \gamma^k > 0$ are the "correlated distortions" as in Bartelsman et al. (2013), δ^y, δ^k captures firm characteristics (such as management quality in Bloom et al. (2019)) and η^y_{it} and η^k_{it} are the uncorrelated part of distortion as in Hsieh and Klenow (2009).

Define $\tilde{\eta}^y_{it} \equiv \delta^y_i + \eta^y_{it}$ and $\tilde{\eta}^k_{it} \equiv \delta^k_i + \eta^k_{it}$. The state vector x_{it} is now defined as

$$x_{it} = [1, k_{i,t-1}, n_{i,t-1}, a_{it}, \tilde{\eta}^k_{it}, \tilde{\eta}^y_{it}]'$$

and the profit function can be approximated as

$$\Pi_{it} \approx \frac{1}{\epsilon}\bar{Y} + \left(1-\frac{1}{\epsilon}\right)\bar{Y}a_{it} - \bar{Y}(\gamma^y a_{it} + \tilde{\eta}^y_{it}) - \bar{R}\bar{K}\left(\gamma^k a_{it} + \tilde{\eta}^k_{it}\right) - \bar{R}\bar{K}\left(\gamma^k a_{it} + \tilde{\eta}^k_{it}\right)k_{it} \\ + \frac{1}{2}\left([\alpha(1-1/\epsilon)-1]\bar{R}\bar{K} - \xi_k\bar{K}\right)k^2_{it} + \frac{1}{2}\left([(1-\alpha)(1-1/\epsilon)-1]\bar{W}\bar{N} - \xi_n\bar{N}\right)n^2_{it} \\ + \frac{1}{2}\left(1-\frac{1}{\epsilon}\right)^2\bar{Y}a^2_{it} - \frac{\xi_k}{2}\bar{K}k^2_{i,t-1} - \frac{\xi_n}{2}\bar{N}n^2_{i,t-1} + \xi_k\bar{K}k_{it}k_{i,t-1} + \xi_n\bar{N}n_{it}n_{i,t-1} \\ + \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}k_{it}n_{it} + \alpha(1-1/\epsilon)^2\bar{Y}k_{it}a_{it} + (1-\alpha)(1-1/\epsilon)^2\bar{Y}n_{it}a_{it} \\ - \left(1-\frac{1}{\epsilon}\right)\bar{Y}(\gamma^y a_{it} + \tilde{\eta}^y_{it})a_{it} - \bar{R}\bar{K}(\gamma^y a_{it} + \tilde{\eta}^y_{it})k_{it} - \bar{W}\bar{N}(\gamma^y a_{it} + \tilde{\eta}^y_{it})n_{it}$$

Everything can be mapped into the primitives in Lemma 1:

$$P = \begin{pmatrix} \frac{1}{\epsilon}\bar{Y} & 0 & 0 & \frac{1}{2}\Big((1-\frac{1}{\epsilon})\bar{Y} - \bar{R}\bar{K}\gamma_k - \bar{Y}\gamma_y\Big) & -\frac{1}{2}\bar{R}\bar{K} & -\frac{1}{2}\bar{Y} \\ * & \frac{1}{2}\bar{R}\bar{K}\Big(\alpha(1-\frac{1}{\epsilon})-1\Big) & \frac{1}{2}\alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} & \frac{\epsilon-1}{2\epsilon}\bar{R}\bar{K} - \frac{1}{2}\bar{R}\bar{K}(\gamma_k+\gamma_y) & -\frac{1}{2}\bar{R}\bar{K} & -\frac{1}{2}\bar{R}\bar{K} \\ * & * & \frac{1}{2}\bar{W}\bar{N}\Big((1-\alpha)(1-\frac{1}{\epsilon})-1\Big) & \frac{\epsilon-1}{2\epsilon}\bar{W}\bar{N} - \frac{1}{2}\bar{W}\bar{N}\gamma_y & 0 & -\frac{1}{2}\bar{W}\bar{N} \\ * & * & * & \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y} - (1-\frac{1}{\epsilon})\bar{Y}\gamma_y & 0 & -\frac{1}{2}(1-\frac{1}{\epsilon})\bar{Y} \\ * & * & * & * & 0 & 0 \\ * & * & * & * & * & 0 \end{pmatrix}$$

$$Q = \begin{pmatrix} 0 \\ \bar{R}\bar{K}\big(\alpha(1-\frac{1}{\epsilon})-1\big) \\ \alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} \\ \frac{\epsilon-1}{\epsilon}\alpha(1-\frac{1}{\epsilon})\bar{Y} - \bar{R}\bar{K}(\gamma_k+\gamma_y) \\ -\bar{R}\bar{K} \\ -\bar{R}\bar{K} \end{pmatrix}, \quad R = \begin{pmatrix} 0 \\ \alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} \\ \bar{W}\bar{N}\big((1-\alpha)(1-\frac{1}{\epsilon})-1\big) \\ \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-\frac{1}{\epsilon})\bar{Y} - \bar{W}\bar{N}\gamma_y \\ 0 \\ -\bar{W}\bar{N} \end{pmatrix}$$

$$H_\iota = \frac{1}{2}\bar{R}\bar{K}\Big(\alpha(1-\frac{1}{\epsilon})-1\Big) - \frac{1}{2}\xi_k\bar{K}, \quad H_{\iota h} = \alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y}, \quad H_h = \frac{1}{2}\bar{W}\bar{N}\Big((1-\alpha)(1-\frac{1}{\epsilon})-1\Big) - \frac{1}{2}\xi_n\bar{N}$$

$$A = \mathrm{diag}(1,1,1,\rho,0,0), \quad B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A.10.3 Private Benefit of Managers

The team production literature (Marschak and Radner (1972)) typically assumes that managers are cooperative and non-strategic: they choose their actions to jointly maximize a common objective, which is the firm's profits. Here I deviate from this assumption and allow capital and labor managers to have private benefits. Following the managerial empire building literature (e.g. Ebsim, Lian, Ma, Ottonello, and Perez (2025)), I model this managerial friction by assuming that the capital manager's payoff function is

$$\Pi_{it} + g(\xi k_{it})$$

where $g(x) = x^{\phi}$ captures the private benefit of the capital manager by diverting a fraction ξ of the capital stock to their career concerns. After doing log-quadratic approximation to the private benefit term g, the capital manager's payoff can be written as

$$x_t' P^k x_t + x_t' Q^k \iota_t + x_t' R h_t + H_\iota^k \iota_t^2 + H_{\iota h} \iota_t h_t + H_h h_t^2$$

where P^k, Q^k and H_ι^k are altered by the private benefit term. The policy function of the capital and labor manager's problem remains linear, and hence Lemma 1 and Lemma 3 continue to hold in this case. The decomposition of investment forecast error in Proposition 1 remains valid.

B Data Appendix

This appendix shows the details of sample construction and variable definition for the expectation test and the calibration exercise.

B.1 Construction of IBES-Compustat Dataset

Prepare the CRSP-Compustat Dataset After downloading the CRSP-Compustat Annual Dataset from the WRDS website, I apply the following filters:

- Keep only the firms incorporated in the US (`fic = "USA"`);
- Keep only the firms using USD as accounting units (`curncd = "USD"`);
- Drop the observations with missing NAICS codes;
- Drop the observations with 2-digit NAICS codes = 52 (finance) and = 22 (utilities);
- Drop the observations with non-positive sales;
- Drop the observations with non-positive employment;
- Drop the observations with missing capital expenditure;
- Drop the observations with missing sales.
- Drop the observations with missing employment.
- Remove duplicate (`LPERMNO`, `fyear`) combinations.

I end up with a CRSP-Compustat Annual sample containing 81,937 firm-year observations from 2000 to 2024. Because the IBES capital-expenditure guidance data used below begin in 2003, I restrict the Compustat comparison sample in Table 1 to the regression sample period, 2003–2024. This period-aligned Compustat sample contains 68,591 firm-year observations from 7,567 firms.

Prepare the IBES-Compustat Joint Sample After downloading the IBES Guidance Dataset from WRDS, I apply the following filters and imputations:

- Keep only U.S. firms reporting in U.S. dollars (`usfirm = 1` and `curr = "USD"`);
- Keep only the annual guidance (`pdicity = "ANN"`);
- Keep capital-expenditure guidance (`measure = "CPX"`). Although the initial data-cleaning code also retains sales guidance, sales guidance is not used in the current expectation regression.
- Keep only observations with range guidance (`range_desc = 1`) or point estimates (`range_desc = 2`). For range guidance, drop observations with a missing lower or upper bound and impute the forecast using the midpoint of the lower and upper bounds, `(val_1 + val_2)/2`.
- Merge the IBES-CRSP bridge file with the IBES sample by matching the IBES ticker and requiring the guidance announcement date to fall between the bridge link's start and end dates. This maps the IBES firm identifier to the CRSP-Compustat identifier `PERMNO`.
- Construct the fiscal-year start and end dates and measure the forecast horizon in quarters. Keep guidance with `cpx_horizon_q < 4`.
- Within each firm and fiscal year, order guidance by its announcement and activation timestamps and retain the first capital-expenditure guidance. A firm-year with a single guidance is retained; intermediate and last-only guidance records are dropped.
- Merge the resulting first-guidance sample with CRSP-Compustat by `LPERMNO` and fiscal year, and retain one observation for each (`LPERMNO`, `fyear`) combination.
- Construct hiring and capital-expenditure lags with exact-year self-joins. In particular, $h_{i,t-1}$ requires employment observations in both $t-1$ and $t-2$.
- For the main Q1 regression sample, keep observations with `cpx_horizon_q = 3`, nonmissing capital-expenditure forecast error, nonmissing lagged hiring, and positive lagged capital expenditure.

The raw IBES file contains no annual U.S.-firm capital-expenditure guidance before fiscal year 2003. Consequently, no observations from 2000–2002 can enter the expectation regression. Although 85 observations in 2003 have a nonmissing capital-expenditure forecast error, only one satisfies the Q1 and lag requirements; the economically substantive sample therefore begins in 2004.

Summary Statistics The final Q1 regression sample contains 14,804 firm-year observations from 2,418 firms and spans 2003–2024. Table 1 compares this sample with the period-aligned CRSP-Compustat sample. Firms in the regression sample are about twice as large in mean sales and employment, while median sales are more than four times larger than in the Compustat sample. The regression sample contains about 21.6 percent as many firm-year observations as the period-aligned Compustat sample. Table B1 reports total sample sales relative to U.S. GDP in selected years. Excluding 2003, which contains only one regression observation, the average annual sales-to-GDP ratio is 29.05 percent over 2004–2024. Because firm sales are gross revenues whereas GDP measures value added, this ratio should not be interpreted as the sample firms' contribution to GDP.

	CRSP-Compustat Sample			
	2005	2010	2015	2020
Total Sales (Billions USD)	8812	10057	11957	13413
Sales/GDP	68%	67%	65%	63%
	IBES/Compustat Sample			
	2005	2010	2015	2020
Total Sales (Billions USD)	2918	5653	5743	5355
Sales/GDP	22%	38%	31%	25%

Table B1: Total Sales Relative to U.S. GDP

Construction of Variables for the Expectation Test The key variables in the expectation tests in Section 3 are constructed as follows. For the capital-expenditure forecast error, I use the arc-percent difference between the forecast and actual values, as in Bloom, Codreanu, and Fletcher (2025):

$$\text{CapEX Forecast Error}_{it} = \frac{\text{CapEX}_{it} - \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}]}{1/2(\text{CapEX}_{it} + \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}])}$$

The arc-difference forecast error is bounded between $[-2, 2]$, limiting the influence of extreme realized or forecast values. It also avoids the problems that arise when taking logs of values close to zero. I verify that the regression results are robust to alternative definitions of the forecast error, including the log difference between realized and forecast capital expenditure.

The definition of hiring is standard. For firm i in year t, hiring h_{it} is defined as

$$h_{it} \equiv \log(\text{emp}_{it}) - \log(\text{emp}_{i,t-1})$$

where `emp` is the employment of the firm in CRSP-Compustat. Alternatively, an arc-difference measure of hiring is widely adopted in the literature as well (e.g. Davis and Haltiwanger (1992)). The regression results do not change with alternative definitions of hiring.

The main regression uses lagged hiring, $h_{i,t-1}$, which requires nonmissing employment in years $t-1$ and $t-2$. Lagged investment is measured as $\log(\text{CapEX}_{i,t-1})$. Both variables are constructed with exact-year lags, and observations with non-positive lagged capital expenditure are excluded before taking logs.

B.2 Construction of the Quantitative Sample for Calibration

The quantitative analysis uses a trimmed sample constructed from the CRSP-Compustat/IBES merged dataset described in Section B.1. The CRSP-Compustat observations underlying this dataset are for U.S.-incorporated firms reporting in U.S. dollars. I require a nonmissing NAICS code, exclude utilities (NAICS 22) and finance (NAICS 52), require positive sales and employment, and require nonmissing capital expenditures. The underlying CRSP-Compustat file spans 2000–2024. Because the IBES capital-expenditure guidance begins in 2003, the final quantitative sample spans 2003–2024.

Value Added and Factor Shares I construct value added and factor shares as follows:

1. Define industries using three-digit NAICS codes for manufacturing and two-digit NAICS codes for other sectors. Merge industry-year wage data from the Census County Business Patterns with the CRSP-Compustat/IBES sample.

2. For observations with missing Compustat labor expense, impute the wage bill as employment multiplied by the corresponding industry-year wage.

3. Define material expenditure as

$$\text{Material Expenditure}_{it} = \text{COGS}_{it} + \text{XSGA}_{it} - \text{DP}_{it} - \text{Wage Bill}_{it}.$$

Drop observations with missing or negative material expenditure.

4. Define value added as sales minus material expenditure and retain observations with positive value added.

5. With $\epsilon = 4$, calculate the aggregate labor and capital shares as

$$s_N = \frac{\sum_{i,t} \text{Wage Bill}_{it}}{\sum_{i,t} \text{Value Added}_{it}}, \qquad \alpha = 1 - \frac{s_N}{1 - 1/\epsilon}, \qquad s_K = \alpha(1 - 1/\epsilon).$$

   The main quantitative analysis uses these aggregate shares. I also construct industry-specific shares as an alternative.

Construction of Quantitative Variables I define log capital and employment as $k_{it} = \log(\text{PPEGT}_{it})$ and $n_{it} = \log(\text{emp}_{it})$. Under the time-to-build timing assumption, current production uses the previous year's capital stock. The value-added productivity residual is therefore

$$a_{it} = \log(\text{Value Added}_{it}) - s_N n_{it} - s_K k_{i,t-1}.$$

The marginal-revenue-product proxies use the same timing convention:

$$mrpk_{it} = \log(\text{Value Added}_{it}) - k_{i,t-1}, \qquad mrpn_{it} = \log(\text{Value Added}_{it}) - n_{it}.$$

I construct investment growth $\Delta k_{it} = k_{it} - k_{i,t-1}$, hiring $h_{it} = n_{it} - n_{i,t-1}$, productivity growth $\Delta a_{it} = a_{it} - a_{i,t-1}$, and the time-to-build log capital–labor ratio $k_{i,t-1} - n_{it}$. All lags are constructed by exact-year firm-level matches, so a missing intermediate year does not generate an incorrect lag.

The IBES component retains the first annual capital-expenditure guidance within the fiscal year with a forecast horizon below four quarters. Unlike the Q1 regression sample, the quantitative sample is not restricted to `cpx_horizon_q = 3` and includes first-guidance horizons from zero to three quarters. Sales guidance and sales-nowcast errors are not used.

Industry-Year Residualization and Trimming I map fiscal years to calendar years by assigning observations with fiscal-year-end months January through May to the previous calendar year and all other observations to the reported fiscal year. I then remove industry-by-calendar-year fixed effects from investment growth, hiring, productivity growth, the productivity level, *mrpk*, *mrpn*, and the capital-labor ratio.

Before trimming, I construct changes in residualized investment growth and hiring using exact one-year lags. I calculate the first and ninety-ninth percentiles in the untrimmed sample and jointly retain observations within these bounds for the following variables:

- the capital-expenditure forecast error, lagged capital expenditure, and lagged hiring;
- changes in residualized investment growth and hiring;
- residualized productivity growth and the residualized productivity level;
- residualized *mrpk*, *mrpn*, and the capital-labor ratio.

This procedure produces a quantitative sample of 13,194 firm-year observations from 2,086 firms over 2003–2024. It is distinct from the Q1 regression sample: 10,261 quantitative-sample observations have a three-quarter forecast horizon, while 2,933 have horizons from zero to two quarters.

Table B2 compares the quantitative sample with the CRSP-Compustat sample before the IBES merge, restricted to the same 2003–2024 period.

Variable	Units	Mean	Std. dev.	Median	Observations
Panel A: Quantitative sample					
Sales	Millions USD	5,346.81	14,803.90	1,755.31	13,194
Gross PPE	Millions USD	3,086.40	6,718.11	894.13	13,194
Capital expenditures	Millions USD	232.89	488.00	69.40	13,194
Employment	Thousands	15.38	32.89	6.00	13,194
Panel B: CRSP-Compustat sample, 2003–2024					
Sales	Millions USD	3,903.97	18,756.48	437.51	68,591
Gross PPE	Millions USD	2,762.10	15,765.38	172.62	65,412
Capital expenditures	Millions USD	230.50	1,382.79	13.20	68,591
Employment	Thousands	11.43	53.99	1.30	68,591

Table B2: Summary Statistics for the Quantitative and Compustat Samples